

Navigating the Commons

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Inefficiencies in the commons arise not only from resource use by existing participants but also from their capacity investment and the entry of new firms. This paper develops a model of firm dynamics with common-pool externalities and estimates it using firm-level panel data from the American whaling industry (1804–1909). I find that ignoring firms' entry, exit, and investment understates the shadow value of whales, so a policy built on it forgoes much of the achievable welfare gain. In the long run, faster technological progress and longer-lasting demand lower welfare under open access, as excessive entry and investment exacerbate externalities.

Keywords: common-pool resources, externalities, firm and industry dynamics, American whaling

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“...whether Leviathan can long endure so wide a chase, and so remorseless a havoc.” — Herman Melville, *Moby-Dick*

1. Introduction

This paper investigates how firm entry, exit, and investment shape the tragedy of the commons by analyzing the 19th-century American whaling industry. Commons problems arise when one agent’s use of a shared resource reduces others’ returns, as in overfishing, deforestation, or groundwater overdraft (Hardin 1968; Ostrom 1990; Stavins 2011; Libecap 2024). While much research focuses on resource use by existing firms, less is known about their capacity investment, their exit, and the entry of new firms. These firm-level dynamics determine how a shared resource is allocated in the long run, and ignoring them may understate the cost of externalities.

Studying firm dynamics in the commons presents three main challenges. First, it requires long panel data that link firms’ entry, exit, and investment to resource dynamics, yet such datasets are rarely available in open-access environments. Second, firm behavior and externalities are connected through complex feedback loops. Each firm’s actions affect stock depletion and congestion, and these effects in turn shape future decisions, complicating empirical analysis. Third, measuring externalities requires accounting for both contemporaneous and future effects across all affected agents. Doing so is especially demanding as industry structure evolves, continually changing how externalities propagate over time.

To address these challenges, I first construct a firm-level panel of American whaling, tracing the industry’s evolution from its rise to its decline. I then develop a dynamic model in which firms enter, exit, and invest in vessels. Firms are heterogeneous in vessel capacity and productivity, making decisions based on their own state and a few aggregates that summarize externalities. This structure makes the dynamic feedback between firm actions and externalities tractable. Finally, I solve the social planner’s problem to recover the shadow value of a whale (and the marginal external cost of harvesting) and quantify the cost of open access over a hundred years.

Three main findings emerge from the estimates and counterfactuals. First, the shadow value of a whale was substantial even compared to the market value of a whale. It ranged from 25 percent of a whale’s market price in 1815 to 57 percent in 1842. A whale taken this year lowers the catch of every voyage that sails later. That future catch depends on which firms enter, which exit, and how much vessel capacity they build over time. A policy that ignores those dynamic decisions delivers only 47 to 78 percent of the gain available from one that accounts for full industry dynamics.

Second, open access overbuilt capacity, dissipating 32 percent of first-best welfare.

At its peak, the industry held 4 times the vessel capacity the first best would ever have reached. The extra vessels further depleted the whale stock and crowded the hunting grounds, so harvesting efficiency fell by 46 percent between the 1820s and the 1850s. On the other hand, under the first best, harvesting efficiency would have risen by 34 percent over the same years.

Third, better technology and longer-lasting demand raise the industry’s returns, yet both lower welfare under open access. A higher return attracts firm entry and vessel investment whose external cost no one internalizes. Doubling the growth rate of whaling technology would have raised the loss to 44 percent of first-best welfare (instead of 32 percent under the actual history). Petroleum was discovered in 1859, substituting for whale products and reducing the demand for them. Had that demand persisted, the welfare loss would have been 63 percent of first-best welfare. In sum, technology growth, which drove the industry’s prosperity, worsened the tragedy of the commons, while the petroleum discovery, which was disastrous for the industry, alleviated the commons problem.

These findings draw on new data, a model of firm dynamics with externalities, structural estimation of its parameters, and counterfactual analyses. I construct a firm-year panel from the expanded American Offshore Whaling Voyages Database (Mystic Seaport Museum, New Bedford Whaling Museum, and Nantucket Historical Association 2024b), supplemented with *Ship Registers* to fill in missing firm information.¹ The sample covers U.S.-registered voyages from 1804 to 1909 and records vessel tonnage, output volumes, and crew size. I link these firm-year observations to annual whale-stock estimates and product prices, yielding a panel that spans the industry’s rise and decline with near-complete coverage of U.S. voyages. At its mid-century peak, when externalities were most severe, American vessels dominated global whaling, accounting for roughly 750 of 900 vessels worldwide (Moment 1957). This setting provides a rare opportunity to study a large-scale common-pool resource in the absence of regulation.

The model combines common-pool externalities with firm and industry dynamics. Whaling firms make entry, exit, and investment decisions based on their vessel capacity and productivity (Hopenhayn 1992; Ericson and Pakes 1995).² Their actions generate externalities they do not internalize: a stock externality, as overexploitation reduces future resource availability (Gordon 1954), and a congestion externality, as contemporaneous

¹I use “firms” and “whaling agents” interchangeably. Section 2.3 details the data construction.

²Productivity is a key source of heterogeneity in whaling. Firm-level panel data show wide dispersion in output. Some firms harvest only a few whales in a year, while others catch more than a hundred (Table 1). Observable characteristics such as vessel capacity and crew size cannot fully explain this persistent, systematic variation.

activity crowds the hunting grounds (Smith 1969). I model production as a function of firms' own states and two aggregates that summarize the externalities. Firms' actions jointly determine these aggregates, which in turn feed back into individual incentives (Huang and Smith 2014). This structure keeps equilibrium computation tractable while capturing externalities in firms' payoffs.

The estimation of the model proceeds in two steps. In the first step, I identify the production function and the demand curve. I estimate the production function using firm-level panel data on physical inputs (vessel capacity and crew size) and output (whales harvested). To address endogeneity between productivity and input choices, I employ a standard dynamic panel approach that models productivity as an AR(1) process and uses lagged inputs as instruments. I estimate the demand curve using supply-side cost shifters (whale stocks five years earlier and long-term bond yields in New England) as instruments to recover the price elasticity consistently.

In the second step, I estimate dynamic costs for entry, exit, and capacity adjustment using a full-solution approach. Estimating these costs requires computing firms' optimal decision rules for both discrete (entry and exit) and continuous (capacity adjustment) decisions, which can be sensitive to the discretization of the continuous state and action spaces. I therefore adopt the recent methodology of Gowrisankaran and Schmidt-Dengler (2025). This approach introduces random shocks to marginal adjustment costs, making the implied choice probabilities robust to the discretization. I then use the implied equilibrium choice probabilities to estimate the dynamic cost parameters that best rationalize firms' observed decisions in the data.

For the counterfactuals, I solve for the allocation a social planner would choose. The planner decides how many firms enter, how many exit, and how much vessel capacity they hold, accounting for the social value of the whales they harvest. A whale left in the ocean reproduces, and it raises what every later voyage harvests, because a larger whale stock makes every firm more productive. Hence, harvesting more whales imposes an additional cost on everyone else, which is the shadow value of a whale. Based on the shadow value, I consider a simple policy designed *with* and *without* firms' entry, exit, and investment to examine the implications of firms' long-run dynamic decisions. Lastly, I investigate how technological progress and demand shifts change firms' equilibrium actions and affect the tragedy of the commons.

Relationship to the literature. The main contribution of this paper is to show how firm-level dynamic decisions shape the long-run allocation of common-pool resources.

My model builds on recent advances in the study of common-pool resources and industry dynamics. A seminal contribution is Huang and Smith (2014), who examine dynamic

strategic interactions in shrimp fisheries. The central challenge in their setting is that agents' payoffs depend on others' actions through dynamic externalities. To address this issue, they assume that fishers affect others only through the aggregate number of vessels, which in turn governs shrimp stocks. Their model features a fixed set of players, which is appropriate over relatively short horizons. Over longer horizons, especially under open access, firms enter, exit, and invest. A large literature on industry dynamics shows that these firm-level decisions shape industry structure and resource allocation (see Aguirregabiria, Collard-Wexler, and Ryan 2021 for a review). I connect these two strands by embedding entry, exit, and capacity adjustment into a common-pool industry equilibrium. With this framework, I highlight how entry and investment can outpace exit and divestment, generating persistent inefficiency through overcapacity.

This paper also relates to a growing literature on management institutions in fisheries. Costello, Gaines, and Lynham (2008) show that catch shares sustain global fish stocks, and Costello et al. (2016) compare overfishing outcomes across management regimes. Researchers have also examined policy effects beyond stock levels. Ho (2023) studies productivity gains from individual transferable quotas. Englander (2023) identifies temporal and spatial spillovers from temporary closures. Aspelund (2025) analyzes redistribution and efficiency effects of trade restrictions in fishing-permit markets. Segerson and Squires (1993) highlight capacity choices, showing that output quotas can induce divestment. A theoretical literature identifies capital as the important margin, because vessels, once built, are long-lived (Clark, Clarke, and Munro 1979). However, direct evidence on the welfare effects of firm dynamics in the commons remains scarce. I model firms' dynamic decisions as determinants of long-run efficiency in commons regulation and quantify the dynamic costs of open access over a full industry life cycle.

A related literature shows how economic environments shape the history of shared resources. Brander and Taylor (1998) model Easter Island's collapse as a renewable-resource feedback, and Taylor (2011) attributes the near-extinction of the American bison to international trade. Copeland and Taylor (2009) show how technology, institutions, and prices shape commons outcomes. Noack and Costello (2024) trace overexploitation to credit expansion without secure property rights. I find that the substitution of petroleum for whale products reduced the cost of open access, while advances in whaling technology raised it.

Beyond the commons, a growing literature studies regulation design in environmental and energy markets with industry dynamics. Fowlie, Reguant, and Ryan (2016) examine market-based environmental regulation under market power and emissions leakage. Elliott (2024) investigates carbon taxes and capacity subsidies when emissions trade off

against intermittency from wind and solar. Aronoff and Rafey (2026) study welfare gains and externalities from environmental offsets. Butters, Dorsey, and Gowrisankaran (2025) analyze the equilibrium effects of large-scale battery storage. Gowrisankaran, Langer, and Zhang (2025) and Chen (2024) show that policy uncertainty shapes dynamic market efficiency. Relative to this literature, I develop an empirical framework for firm dynamics in the commons, in an environment where the externalities complicate the dynamic problem. The framework of this paper applies to many industries whose central distortion comes from a shared resource.

Finally, this paper speaks to the literature on long-run productivity in economic history, which has traced productivity growth to management practices (Braguinsky et al. 2015; Giorcelli 2019; Bianchi and Giorcelli 2022; Rubens 2023), to technology adoption and innovation (Hanlon 2015; Braguinsky et al. 2021; Juhász, Squicciarini, and Voigtländer 2024; Hornbeck et al. 2025), and to industrial policy (Juhász 2018; Lane 2025). In American whaling, harvesting efficiency falls while the industry grows, even as whaling technology improves, because every firm draws down the shared whale stock. A decades-long productivity decline in a shared-resource industry is a feature unique to open access, and the counterfactuals show how much of that decline management would have restored.

Outline. The rest of the paper is organized as follows. Section 2 provides a historical overview and descriptive analyses of the American whaling industry. Section 3 develops the model of common-pool industry dynamics. Section 4 describes the estimation strategies and presents the results. Section 5 implements counterfactual exercises. Section 6 concludes.

2. The American whaling industry

This section provides a historical overview of the U.S. whaling industry and describes the data. I document the industry’s major products and the shifts in market size (Section 2.1), the primary decision makers (Section 2.2), data sources (Section 2.3), and descriptive statistics (Section 2.4).

2.1. Products and markets

The American whaling industry primarily produced three commodities: sperm oil, whale oil, and whalebone (baleen). Sperm oil, rendered by boiling the blubber of sperm whales, was prized for its clean, bright flame and was used for illumination and as a lubricant for fast-moving machinery (e.g., spindles). Whale oil, extracted from baleen whales, was

used as a lubricant for heavy machinery and as an illuminant. Whalebone, taken from the mouths of baleen whales, was used in products such as corsets. Sperm whales were hunted on warm-water grounds across all oceans, while the baleen species, chiefly right and bowhead whales, concentrated in higher latitudes (Tillman and Donovan 1983). Processing took place on board. Blubber was rendered in tryworks, brick furnaces built into the deck, which let vessels stow oil and remain at sea for years. The product market consisted of homogeneous goods and was competitive. Individual whaling firms had little ability to affect prices or differentiate their outputs from those of other firms (Davis, Gallman, and Gleiter 2007).

During the first half of the nineteenth century, industrialization in the United States substantially increased demand for oil, expanding markets for whaling. During this period, the United States overtook Britain as the leading whaling nation, supported by strong domestic demand growth, technological and institutional advantages, and higher-skilled crews (Davis, Gallman, and Hutchins 1987). The industry entered its “Golden Age” during the 1830s–50s, when American whalers produced over 90% of the world’s sperm oil (Tillman and Donovan 1983).³ The mid-1840s brought a series of shocks. Coal oil emerged as a credible substitute, the California gold rush drew away crews, and voyages shifted increasingly toward Pacific grounds, which lengthened voyage duration. Thereafter, the discovery of petroleum in 1859 and the disruption of the Civil War (1861–1865) prompted the decline of the industry.⁴ By the late nineteenth century, output and value had fallen back to levels comparable to those of the early 1800s.

2.2. Whaling agents

Whaling agents, interchangeably referred to as “firms” in this paper, were the primary decision makers (see Chapter 10 of Davis, Gallman, and Gleiter 2007 and Chapter 1 of Nicholas 2019 for a more in-depth discussion of whaling agents). The median firm ran two vessels in a given year, and roughly half of firm-year observations involved a single vessel (Table 1). As American whaling voyages spanned multiple seasons (typically two to three

³I treat non-American whaling operations as exogenous given their relatively small scale. If they responded endogenously to counterfactual regulations on American whaling but were not themselves regulated, the effects of the planner’s management in Section 5 would be smaller.

⁴Whalers understood the threat quickly. The *Whalemen’s Shipping List and Merchants’ Transcript* (WSL, July 2, 1861) reported:

At New Bedford the business has declined about one-third during the past three years, and it is believed will decline fully a third more within the present year [1861] . . . Oil, which costs 60 cents to produce, will now bring but 40 cents.

Likewise, Tower (1907) dates the fate of the whale fishery to the opening of the first Pennsylvania oil well.

years), success depended on effective planning and management by agents. They were responsible for acquiring vessels and hiring a captain, as well as determining the necessary equipment and crew. In the process of decision making, agents utilized logbooks from prior voyages. These contained daily notes made by captains and first mates, providing detailed information on whale sightings, weather, locations, and crew morale. At the end of a venture, the logbook became the property of the agent, serving as a repository of accumulated knowledge (Nicholas 2019).

Plans by agents included voyage duration, hunting grounds, locations and dates for resupplying, and shipping oil or bone back home (see Online Appendix B for an example). The plan was made before the vessel sailed but was frequently subject to change. Unforeseen success or challenges could necessitate altering the original schedule. For instance, exceptional success might prompt an early stop at a transshipment point; setbacks could extend the time at sea beyond the initial plan. While the captain made day-to-day sailing and hunting decisions, the agents held general authority and communicated with the captain as consistently as they could. Before the voyage commenced, the captain and agent agreed on dates and stations for exchanging letters. These exchanges occurred at predetermined rendezvous points, either between whaling vessels or between supply ships and whaling vessels (see Online Appendix B for an example).⁵

In many cases, whaling agents owned vessels as principal owners, flying a “house flag” that signaled ownership. But this did not mean the agent alone financed the voyage. Multiple investors typically contributed, with arrangements varying from one voyage to another (see Online Appendix Figure C2 for an example). The reliance on multiple investors reflected the substantial capital required to outfit a whaling expedition. A typical New Bedford voyage in the 1850s cost between \$20,000 and \$30,000 (equivalent to roughly \$831,000–\$1,200,000 today). The cost was considerable, especially when compared to other sectors. The value of the average manufacturing firm’s capital stock was about \$4,335 in 1850 ($\approx$ \$180,000 today) and \$7,191 in 1860 ($\approx$ \$285,000 today) (Davis, Gallman, and Gleiter 2007).

2.3. Data

I construct several datasets to examine the whaling market. This subsection describes the new aspects of the data, emphasizing how they reveal (i) firm-level behaviors; (ii) price per whale; (iii) common-pool features; and (iv) demand shifters. See Online Appendix C

⁵For instance, Santa Maria, a small island in the Galapagos archipelago, became a mail hub because it was frequented by vessels hunting whales in the Pacific (Nicholas 2019).

for a detailed description.

The primary source is the American Offshore Whaling Voyages Database (*Voyage Database*), which covers nearly all U.S. whaling voyages undertaken in the nineteenth century (Mystic Seaport Museum, New Bedford Whaling Museum, and Nantucket Historical Association 2024b).⁶ The database records vessel characteristics (name, unique ID, tonnage, rigging), the managing owner/agent, captains, outbound and inbound years, outputs (sperm oil, whale oil, whalebone), ports of departure and return, and other related information (see Online Appendix Figure C1 for an example).

The *Voyage Database* does not include crew information, so I supplement it with the American Offshore Whaling Crew Lists Database (*Crew Lists Database*) to obtain crew size for each voyage (Mystic Seaport Museum, New Bedford Whaling Museum, and Nantucket Historical Association 2024a). The *Crew Lists Database* covers 40 percent of the voyages in the *Voyage Database*.

To study firms' production, entry, exit, and capital accumulation, I build a firm-year panel from voyage records. This construction becomes possible because the *Voyage Database* recently expanded whaling agent coverage from about one-fifth to 67 percent of the 15,707 voyages.⁷ Still, the *Voyage Database* lists no whaling agent for 5,182 of the 15,707 voyages, so I further supplement agent information by manually checking the data and reading port *Ship Registers* (see Online Appendix C.2). Together these raise agent coverage to 12,613, or about 80 percent of all voyages.

Based on this expanded dataset, I divide each voyage's output by its duration (measured from departure to return) and aggregate to the firm-year level by whaling agent. I convert the three outputs (sperm oil, whale oil, and whalebone) into the *number of whales harvested* using historical benchmarks (see Online Appendix C.4). I also aggregate voyages' inputs (vessel tonnage and crew size) to the firm-year level, assuming they remain constant throughout each voyage.⁸

The prices of whaling products come from Tower (1907) and Davis, Gallman, and Gleiter (2007). I calculate the price *per whale* as the aggregate value of outputs divided by the total number of whales harvested each year. The aggregate output value equals sperm oil quantity $\times$ sperm oil price + whale oil quantity $\times$ whale oil price + whalebone quantity $\times$ whalebone price. I compute the total number of whales harvested by summing catch

⁶Prior studies in economics use related data sets to study risk sharing in financial markets, firm formation and corporate organization, productivity, and labor diversity (Hilt 2006, 2007, 2008; Davis, Gallman, and Gleiter 2007; Baggio and Cosgel 2024).

⁷Earlier studies relied on substantially more limited coverage (e.g., Hilt 2006; Hilt 2007; Hilt 2008; Davis, Gallman, and Gleiter 2007).

⁸Although crew desertion and recruitment occurred, vessel tonnage largely constrained crew size.

TABLE 1. Firm-year level statistics for the American whaling industry

Variable	Unit	Mean	SD	P5	Median	P95
Panel A. Outputs						
Whales harvested	Number	41.859	50.396	1.99	23.69	146.23
Sperm oil	Thousand barrels	0.527	0.729	0.00	0.28	1.90
Whale oil	Thousand barrels	0.802	1.322	0.00	0.23	3.60
Whale bone	Thousand pounds	6.927	14.174	0.00	0.15	34.05
Panel B. Inputs						
Vessel capacity	Tonnage	767.285	822.289	105	396	2,501
	Number	2.665	2.445	1	2	8
Crew size	Number	85.766	85.208	16	54	268
Firm age	Year	12.739	12.122	1	9	38
Panel C. Dynamic decisions						
Entry	Rate	0.103	0.098	0.000	0.089	0.305
Exit	Rate	0.099	0.057	0.000	0.091	0.199
Investment	Rate	0.147	0.112	0.009	0.135	0.273
	Tonnage	321.695	209.904	67.0	306.8	730.7
Divestment	Rate	0.137	0.081	0.000	0.143	0.269
	Tonnage	-340.655	237.981	-786.0	-309.0	-71.8

Note: The dataset includes 10,341 firm-year observations and 965 firms, covering the years 1804 to 1909. Reported quantities of sperm oil, whale oil, and whalebone determine the number of harvested whales (see Online Appendix C.4 for details). Crew size is observed for 5,699 firm-year observations because the *Crew Lists Database* is a subsample of the *Voyage Database*. Entry and exit rates are defined as the fraction of firms that enter or exit in a given year; investment and divestment rates are defined analogously. Changes in vessel tonnage between the current year and the following year capture investment and divestment, excluding changes due to wreckage.

records across all voyages. Online Appendix Figure E2 reports the annual whale price.

Two aggregate series complete the panel. Total vessel tonnage, summed over all whaling vessels in each year from the *Voyage Database*, measures industry-wide capacity. The whale stock series is projected from whale population estimates (Davis, Gallman, and Hutchins 1988; Whitehead 2002) and catch data using a discrete-time generalized logistic model, described in Section 3.1.

Finally, I use two exogenous variables to capture demand shifters: U.S. GDP per capita and petroleum prices. These data come from the Historical Statistics of the United States (Carter et al. 2006).⁹

⁹The online version is available at <https://hsus.cambridge.org/>. Annual U.S. GDP per capita is from Series Ca9, and petroleum prices are from Series Db56. Online Appendix Table E1 reports summary statistics.

2.4. Descriptive statistics

Table 1 reports firm-year statistics that highlight substantial variation across observations. Panel A shows that some firms harvested only a few whales, while others harvested a hundred or more. Panel B shows wide dispersion in operational scale. Annual operations ranged from a single vessel to 8 or more, with corresponding variation in vessel tonnage and crew size. Firm lifetimes also varied substantially. Some firms operated only briefly, while others remained for decades. The median firm spent 6 years in the industry, equivalent to two to three voyages at typical voyage lengths.¹⁰ Panel C summarizes dynamic decisions. Entry is defined as the first appearance as a whaling agent in the dataset, and exit means leaving whaling altogether. Capacity adjustment occurred at the vessel margin: firms invested by acquiring additional or larger vessels and divested by selling or retiring them.¹¹ Investment and divestment are measured as changes in vessel tonnage from year t to year $t + 1$, excluding changes due to wreckage.¹² On average, relative to the number of firms active in each year, 10% entered, 10% exited, 15% invested, and 14% divested. Median investment was about 307 tons and median divestment about 309 tons, with substantial dispersion in both, reflecting the lumpy and uncertain nature of capacity adjustment.

Figure 1 presents the evolution of total whaling vessel capacity and whale stock. Panel A shows the rise of American whaling to global dominance during the industry's Golden Age, followed by its collapse within a single century. In 1816–1819, the industry's capital stock averaged only 19 thousand vessel-tons a year. Over the subsequent three decades, total tonnage increased more than 12-fold, peaking at 233,400 tons in 1845. Panel B displays the projected whale population, the sum of the baleen and sperm whale stocks (see Section 3.1).¹³ Whale stocks experienced sharp declines during the Golden Age, coinciding with the rapid expansion of vessel capacity and hunting pressure.

Figure 2 provides suggestive evidence of negative externalities. Panel A displays the

¹⁰The firm-age row of Table 1 reports age at each firm-year observation, whose median (9 years) exceeds the median firm's 6-year span because long-lived firms contribute more observations.

¹¹For example, the *Addison*, built at Philadelphia in 1816, sailed as a merchant ship for seventeen years. In 1833 the Fairhaven agents Gibbs & Jenney bought her for whaling and sent her on two voyages. In 1838 she passed to New Bedford, where A. H. Seabury and then Isaac B. Richmond ran her on seven voyages through 1867. At the end of that voyage she was sold at auction; Lorenzo Peirce bought the largest share and, as her agent, sent her on three more voyages from New York and New Bedford. In 1874 she was sold "to be employed in the merchant service" (Davis, Gallman, and Gleiter 2007; agents from the *Voyage Database*).

¹²Shipwrecks were a constant hazard. Roughly 7 vessels per year were destroyed over the sample period, with spikes during the Civil War and the Arctic disaster of 1871; about 7% of firm-year observations record a wreck. In the model, I treat wrecks as an exogenous destruction hazard, separate from firms' voluntary decisions.

¹³I treat the whale stock as a single global pool. Whales migrate across ocean basins, American vessels hunted the major grounds worldwide, and no ground-level stock series exists for the period.

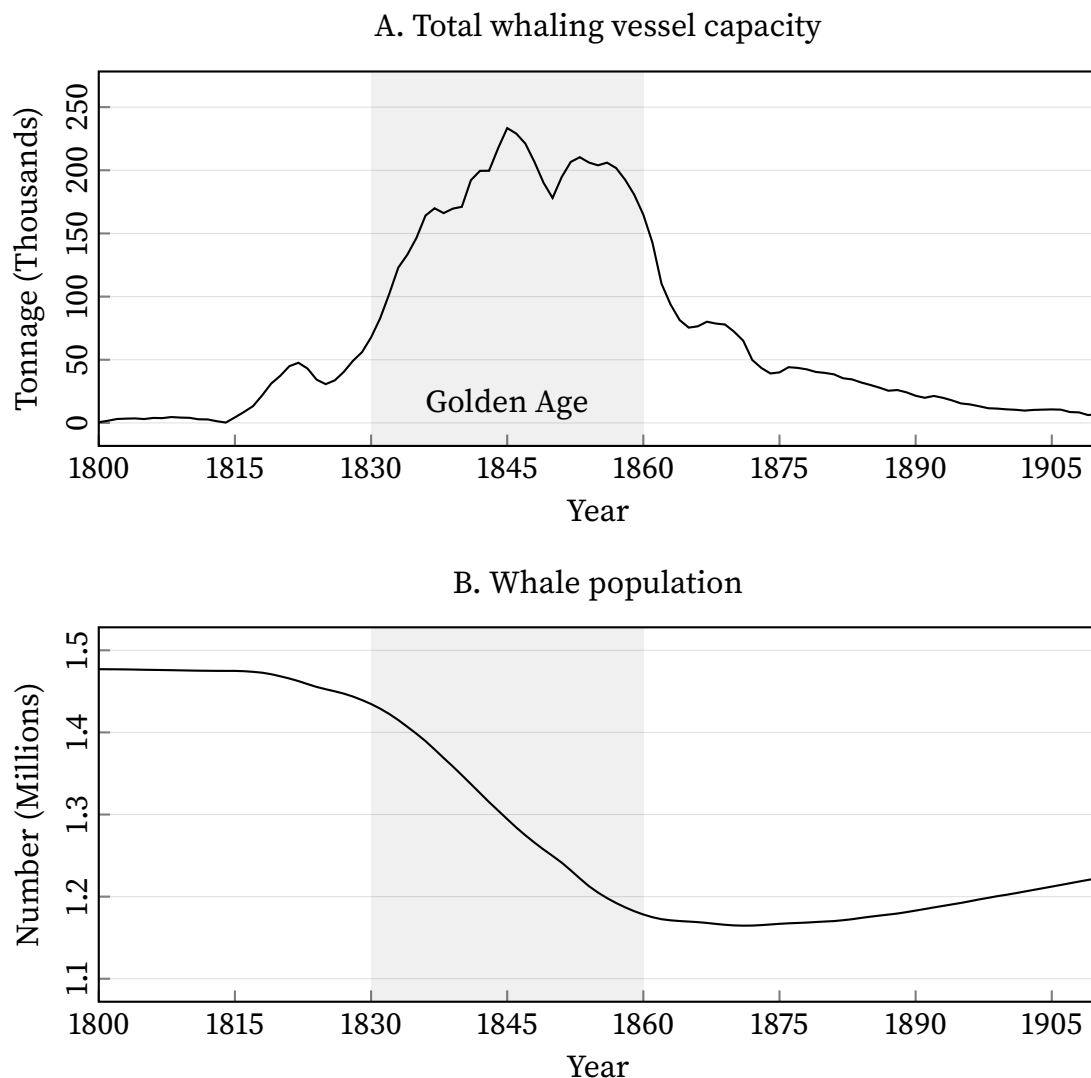


FIGURE 1. Aggregate whaling vessel capacity and whale stock

Note: Total whaling vessel capacity is calculated by summing the tonnage of all whaling vessels for each year from the *Voyage Database*. Whale population is projected using a discrete time, generalized logistic model of whale population dynamics (see Eq. 1 in Section 3.1). The term “Golden Age” refers to the period when the American whaling industry was at its peak, from the 1830s to the 1850s.

number of whales harvested per thousand tons of vessel capacity. Following a marked increase in the 1800s–1820s, it peaked at 87 in 1827 and fell to 44 by the mid-1850s. This decline is striking in light of technological growth over the period, about 1.3% per year from the production function estimates (see Section 4.1). On the other hand, Panel B shows non-decreasing revenue per ton of vessel capacity. It averaged \$31 per ton (in 1880 dollars) in the 1820s and rose to \$48 in the 1850s. A significant rise in prices drove this increase. For instance, the price of sperm oil averaged \$0.53 per gallon (in 1880 dollars) in the 1820s and nearly tripled by the 1850s (see Online Appendix Figure E1). These price increases

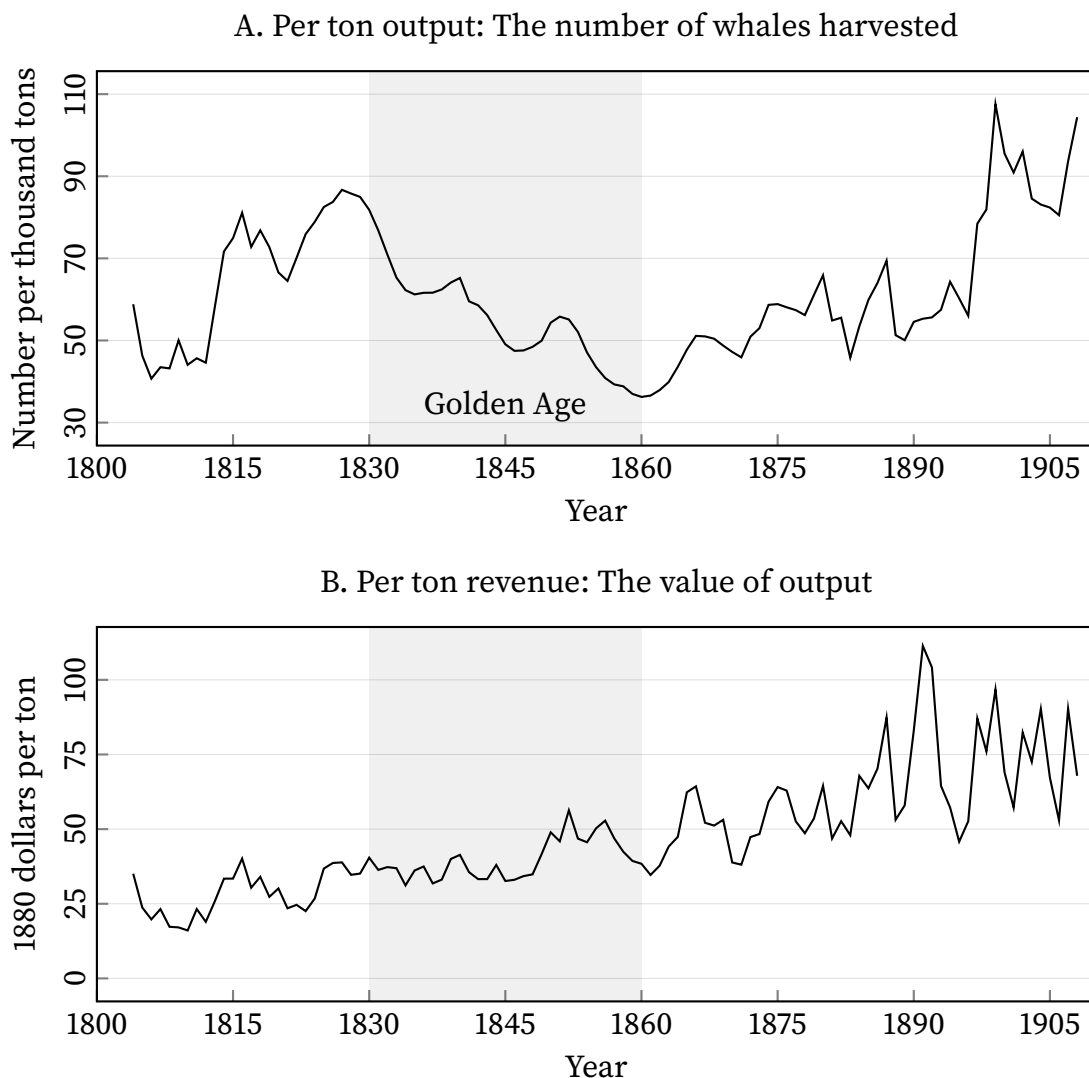


FIGURE 2. Suggestive evidence of negative externalities

Note: In panel A, output is measured by the number of whales harvested per thousand tons of aggregate vessel capacity. In panel B, the value of output per ton of aggregate vessel capacity is calculated as the sum of the product values: sperm oil quantity $\times$ sperm oil price + whale oil quantity $\times$ whale oil price + whalebone quantity $\times$ whalebone price. The term “Golden Age” refers to the period when the American whaling industry was at its peak, from the 1830s to the 1850s.

sustained firms’ incentives to continue operating despite rising costs, reinforcing the equilibrium effect of continued entry and investment amid worsening externalities.

3. A model of common-pool industry dynamics

This section develops a model of competitive industry dynamics for a common-pool resource. The model builds on a large number of price-taking firms (Hopenhayn 1992),

extending to an environment with an endogenous, evolving production externality. It draws on tractable solution methods for industry dynamics with many firms (Weintraub, Benkard, and Van Roy 2008; Benkard, Jeziorski, and Weintraub 2024). The subsections present the model setup (Section 3.1), industry states, beliefs, and decision rules (Section 3.2), dynamic optimization (Section 3.3), and the equilibrium (Section 3.4).

3.1. Setup

Time. Each period corresponds to one year and is indexed by $t \in \{1, 2, \dots, T\} \subset \mathbb{N}$, where T denotes the terminal year.¹⁴ A finite horizon is crucial for the empirical analysis because it allows the model to capture the industry’s rise and fall in a nonstationary environment.¹⁵

Firms. Firms are indexed by i . In year t , the set of incumbents is $\mathcal{J}_t$, with $|\mathcal{J}_t| = N_t$, and the set of potential entrants is $\mathcal{J}_t^{\text{pe}}$, with $|\mathcal{J}_t^{\text{pe}}| = N_t^{\text{pe}}$.¹⁶

Resource. The whale stock in year t is denoted by $W_t \in [0, W_1]$, where W_1 is the natural stock level before the large-scale harvesting by American whalers.¹⁷ The stock evolves according to a discrete-time generalized logistic model of population dynamics (Whitehead 2002; Baker and Clapham 2004; Breiwick and York 2009), net of the year’s harvest:

$$(1) \quad W_{t+1} = \left(W_t + rW_t \left[1 - \left(\frac{W_t}{W_1} \right)^z \right] - Q_t \right) \phi_{t+1}^W,$$

where r is the whale-stock regeneration rate, z is the exponent setting the maximum sustainable yield level (MSYL),¹⁸ Q_t is the total whale harvest in year t , and ϕ_{t+1}^W is the random shock to the whale stock with $\log \phi_{t+1}^W \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_W, \sigma_W^2)$. There was no contemporaneous assessment of whale abundance, so each firm acts on the distribution of next year’s stock based on its beliefs. I write $\tilde{W}_{t+1}$ for the stock under that distribution, whose dispersion is

¹⁴For empirical analysis, I set T to 1915 because the American whaling industry was non-existent by then.

¹⁵By contrast, the infinite-horizon stationary framework does not naturally accommodate this historical pattern.

¹⁶The pool of potential entrants N_t^{pe} is set from the data. In each year it is 1.75 times the largest entry count observed over the preceding ten years. Though alternative specifications of the potential-entrant pool affect dynamic cost estimates, the findings from counterfactuals are robust.

¹⁷ $W_1 \approx 1.48$ million whales, comprising about 1.11 million sperm whales (Whitehead 2002) and about 367,000 baleen whales (Davis, Gallman, and Hutchins 1988). I project the two populations forward from these pre-exploitation levels under the generalized-logistic law (1), net of recorded catch, to obtain the stock series in Figure 1.

¹⁸The MSYL is the stock level at which the maximum number of whales can be taken without changing the stock level.

governed by σ_W .¹⁹

Firm state. Active whaling firms are heterogeneous along two dimensions. First, capacity (K_{it}) captures the tonnage of operating vessels. Second, productivity (Ω_{it}) reflects managerial efficiency, modeled as Hicks-neutral technology.²⁰ The *individual* state of firm i in year t is denoted by $x_{it} \equiv (K_{it}, \Omega_{it}) \in \mathcal{X}$. Vessel capacity changes through costly adjustment (investment or divestment) I_{it} :

$$(2) \quad K_{it+1} = (1 - \delta)K_{it} + I_{it},$$

where δ is the vessel depreciation rate. The index of (log) productivity, $\omega_{it} = \log \Omega_{it}$, follows a first-order Markov process:

$$(3) \quad \omega_{it+1} = \mathbb{E}[\omega_{it+1} | \omega_{it}] + v_{it+1},$$

where v_{it+1} is the unexpected productivity innovation, with $\mathbb{E}[v_{it+1} | \omega_{it}] = 0$. Potential entrants are identical prior to entry, sharing a common entry state x^e .²¹

Production. The whale-harvesting function $\mathcal{H}$ gives the number of whales harvested by firm i in year t :

$$(4) \quad Q_{it} = \mathcal{H}_t(x_{it}, W_t, K_t; \beta).$$

The harvesting function incorporates the stock externality through W_t and the congestion externality through aggregate whaling vessel capacity K_t (the more vessels, the more congested the hunting grounds). Thus, firm i 's harvest depends not only on its own state (x_{it}) but also on these external factors. The time subscript in $\mathcal{H}_t(\cdot)$ reflects exogenous technological growth in the industry. The vector β contains the parameters governing the harvesting function.

Demand. The inverse whale-demand function $\mathcal{P}$ describes the relationship between price, aggregate harvest, and demand conditions:

$$(5) \quad P_t = \mathcal{P}(Q_t; Z_t^d, \alpha) = \mathcal{P}_t(Q_t; \alpha).$$

¹⁹I set the location μ_W so that the shock leaves expected harvest unchanged.

²⁰Incorporating firm-level, time-varying productivity into the state space addresses the limitation of relying solely on i.i.d. random shocks as the source of unobserved variation over time (Bajari, Benkard, and Levin 2007). It is also crucial for capturing dispersion in output even after accounting for observed firm characteristics such as capacity and crew size (see Table 1).

²¹I set the entrants' capacity and productivity to those of the median entrant observed in the data.

Here, P_t denotes the market price in year t , common to all firms because the American whaling industry was competitive (Davis, Gallman, and Gleiter 2007). The vector Z_t^d consists of exogenous demand shifters, namely U.S. GDP per capita, a time trend, and petroleum prices (since 1859). For notational simplicity, these shifters are absorbed into the time subscript of $\mathcal{P}_t(\cdot)$. The vector α contains the parameters of the demand function.

Period payoff. Firm i 's profit Π in year t is revenue minus labor costs:

$$(6) \quad \Pi_t(x_{it}, W_t, K_t, Q_t; \alpha, \beta) = \mathcal{P}_t(Q_t; \alpha) \mathcal{H}_t(x_{it}, W_t, K_t; \beta) - w_t^L L_{it},$$

where L_{it} denotes crew size and w_t^L is the common wage index for crew. The payoff depends on the decisions of all firms, as reflected in W_t , K_t , and Q_t .

Dynamic actions. In each year t , incumbent firm i chooses whether to exit or remain active. If it stays, the firm may adjust its vessel capacity for the next year. Incumbent actions are denoted $a_{it} \in \mathcal{A} = \{0\} \cup \mathcal{K}$, where 0 represents exit and $\mathcal{K}$ is the set of feasible capacity levels. Potential entrants decide whether to enter the market, with actions denoted $a_{it}^{\text{pe}} \in \mathcal{A}^{\text{pe}} = \{0, 1\}$, where 0 indicates no entry and 1 indicates entry.

Timing. At the start of year t , incumbent firms hunt whales and earn period payoffs as defined in equation (6). After receiving payoffs, each incumbent draws an exit value (ζ_{it}), and each potential entrant draws an entry cost (κ_{it}). An incumbent that stays then draws a marginal capacity adjustment cost (ϵ_{it}) before choosing next year's capacity. Firms observe these draws before deciding, so each shock enters the payoff of its own action.²² At the end of the year, incumbent firms that chose to stay face the wreck hazard. Surviving firms' decisions are implemented, entrants enter, and states transition accordingly.

3.2. Industry states, beliefs, and decision rules

There are many firms, and each is small relative to the industry size. Firms take the market price and the evolution of the industry aggregates as given. They perceive that these aggregates evolve deterministically as idiosyncratic shocks average out across a large number of firms (Weintraub, Benkard, and Van Roy 2008, 2010).²³ Each firm therefore conditions its decisions on its own state, the whale stock, and the industry aggregates.

²²Random shocks introduce ex ante uncertainty into firms' decisions. They help capture variability in the data that would otherwise be hard to match empirically (Rust 1987).

²³Weintraub, Benkard, and Van Roy (2008) introduce their oblivious equilibrium as an approximation to the Markov perfect equilibrium. Their solution concept fits my framework naturally, but I assume a competitive market, not imperfect competition. In the data, the largest firm's share of the annual whale harvest averaged 3.4% over the Golden Age (1830–1860) and never exceeded 4.8%.

As shown in equation (6), the industry aggregates that affect firms' payoffs are total vessel capacity, total whale harvest, and whale stock. Total vessel capacity equals the sum of all firms' capacities:

$$(7) \quad K_t = \sum_{i \in \mathcal{J}_t} K_{it}.$$

Total whale harvest equals the sum of all firms' harvests:

$$(8) \quad Q_t = \mathcal{Q}_t(W_t, K_t; \beta) = \sum_{i \in \mathcal{J}_t} \mathcal{H}_t(x_{it}, W_t, K_t; \beta),$$

which depends on current whale stock and total vessel capacity. Given the initial stock W_1 , the stock's law of motion (1), and the aggregate harvest (8), firms form beliefs about the future path of the whale stock. For notational convenience, I combine the firm state, resource stock, and industry aggregates into a payoff-relevant state $s_{it} = (x_{it}, W_t, K_t, Q_t) \in \mathcal{S}$.

With i.i.d. entry costs, exit values, and capacity adjustment costs, incumbents follow the decision rule $\psi_t(s_{it}, \zeta_{it}, \epsilon_{it}) = a_{it}$, while potential entrants use $\psi_t(s_{it}, \kappa_{it}) = a_{it}^{\text{pe}}$. Here, the time subscript in $\psi_t(\cdot)$ indicates that decision rules are nonstationary due to the evolution of the industry, embodied in exogenous demand shifters and industry-wide technology growth. The year- t profile of decision rules is

$$(9) \quad \psi_t = \{ \psi_t(s_{it}, \zeta_{it}, \epsilon_{it}) \}_{i \in \mathcal{J}_t} \cup \{ \psi_t(s_{it}, \kappa_{it}) \}_{i \in \mathcal{J}_t^{\text{pe}}}.$$

Let $f_t(x)$ denote the number of firms in state x in year t , and let $M_t(x' | x, \psi_t)$ denote the probability that a firm in state x in year t moves to state x' in year $t + 1$ under the decision rules ψ_t . The distribution of firms evolves according to:

$$(10) \quad f_{t+1}(x') = \begin{cases} \sum_{x \in \mathcal{X}} M_t(x' | x, \psi_t) f_t(x) + \sum_{i \in \mathcal{J}_t^{\text{pe}}} \psi_t(s_{it}, \kappa_{it}) & \text{if } x' = x^{\text{e}} \\ \sum_{x \in \mathcal{X}} M_t(x' | x, \psi_t) f_t(x) & \text{otherwise,} \end{cases}$$

where x^{e} is the state of a new entrant. The evolution of the firm distribution depends on the initial distribution f_1 and the initial resource stock W_1 . For notational simplicity, I suppress the year-1 dependence in the remainder of the analysis.

3.3. Dynamic optimization

Firms make their dynamic choices of entry, exit, and capacity adjustment to maximize their expected discounted stream of profits. They discount future profits by a factor $\rho \in (0, 1)$.

A potential entrant draws an entry cost κ_{it} from an i.i.d. logistic distribution with mean $\bar{\kappa}$ and scale σ_L . This entry cost represents a barrier to profitable entry, capturing heterogeneity in firms' ability to successfully enter and operate in the industry.

An incumbent firm faces two shocks. First, it draws an exit value ζ_{it} from an i.i.d. logistic distribution with mean $\bar{\zeta}$ and scale σ_L . This exit value reflects the opportunity cost of remaining in the whaling industry, including outside opportunities. Second, it draws a marginal capacity adjustment cost ϵ_{it} from an i.i.d. normal distribution with standard deviation σ .²⁴ Capacity adjustments involve buying or selling vessels and the related operations. Variation in organizational frictions, financing conditions, and firm-specific constraints generates heterogeneity in adjustment costs. I assume that the three shocks (entry cost, exit value, and marginal adjustment cost) are drawn independently of one another and over time.

In addition to random shocks, incumbent firms incur *deterministic* adjustment costs. These costs depend on current capacity, the magnitude of the adjustment, and whether the firm invests or divests:

$$(11) \quad C(K_{it+1}, K_{it}; \gamma) = \mathbb{1}\{K_{it+1} > K_{it}\} (\gamma_0^+ + \gamma_1^+ I_{it} + \gamma_2^+ I_{it}^2) \\ + \mathbb{1}\{K_{it+1} < K_{it}\} (\gamma_0^- + \gamma_1^- I_{it} + \gamma_2^- I_{it}^2),$$

where $\gamma = \{\gamma_0^+, \gamma_1^+, \gamma_2^+, \gamma_0^-, \gamma_1^-, \gamma_2^-\}$ and I_{it} is defined by equation (2). The first part, indicated by $\mathbb{1}\{K_{it+1} > K_{it}\}$, captures investment costs, whereas the second part, indicated by $\mathbb{1}\{K_{it+1} < K_{it}\}$, represents divestment costs. Each part comprises three elements: a fixed cost (γ_0), a linear cost (γ_1), and a convex cost (γ_2).

An active firm with state $s_{it} = (x_{it}, W_t, K_t, Q_t)$ in year t , given that all firms follow the decision rules ψ_t , solves the following dynamic problem:

$$(12) \quad V_t(s_{it}, \psi_t; \Theta) = \Pi_t(s_{it}; \alpha, \beta) + \mathbb{E}_\zeta \left[\max \left\{ \zeta_{it}, \right. \right.$$

²⁴Stochastic adjustment costs date back at least to Blundell et al. (1992). Introducing stochasticity into *marginal* adjustment costs follows Kalouptsi (2018), Caoui (2023), Gowrisankaran, Langer, and Reguant (2026), and Gowrisankaran and Schmidt-Dengler (2025). In contrast, Ryan (2012) and Fowlie, Reguant, and Ryan (2016) apply stochastic shocks to *fixed* adjustment costs.

$$\mathbb{E}_\epsilon \left[\max_{K_{it+1}} \left\{ -C(K_{it+1}, K_{it}; \gamma) - I_{it} \epsilon_{it} + \rho(1 - p_t) \mathbb{E}_t [V_{t+1}(s_{it+1}, \psi_{t+1}; \Theta) | s_{it}, \psi_t] \right\} \middle| s_{it} \right] \bigg| s_{it},$$

where $\Theta = \{\alpha, \beta, \bar{\zeta}, \bar{\kappa}, \sigma_L, \sigma, \gamma\}$, p_t is the exogenous end-of-period wreck hazard,²⁵ and $s_{it+1} = (x_{it+1}, \tilde{W}_{t+1}, K_{t+1}, Q_{t+1})$.²⁶ Because the expectations over ζ_{it} and ϵ_{it} are taken before those shocks realize, $V_t(\cdot)$ is an ex-ante (integrated) value function. Its time subscript reflects the nonstationarity, as period profits $\Pi_t(\cdot)$ vary over time due to demand shifts and technological progress. A firm exits in year t if and only if its exit value (ζ_{it}) exceeds the expected continuation value. If it stays (and survives wreck-hazard risk), the firm selects the next period's capacity level K_{it+1} to maximize its discounted expected future value net of adjustment costs. The conditional expectation $\mathbb{E}_t[\cdot | s_{it}, \psi_t]$ averages over the firm's productivity ω_{it+1} from (3) and the uncertain stock level $\tilde{W}_{t+1}$ from (1), while the aggregates K_{t+1} and Q_{t+1} are perceived to evolve deterministically.

For a potential entrant, the objective is to solve the following problem:

$$(13) \quad \mathbb{E}_\kappa \left[\max \left\{ 0, \rho \mathbb{E}_t [V_{t+1}(s_{it+1}, \psi_{t+1}; \Theta) | s_{it}, \psi_t] - \kappa_{it} \right\} \middle| s_{it} \right].$$

A potential entrant decides to enter in year t if and only if the discounted expected value net of the entry cost is greater than zero.²⁷

3.4. Equilibrium

The model captures the rise and fall of the American whaling industry, together with shifts in technology, demand, and whale stocks over time. Given this nonstationarity, I solve for equilibrium via backward induction, assuming that the nonstationary equilibrium converges to a stationary equilibrium in the long run (Benkard, Jeziorski, and Weintraub 2024). Because the industry was almost non-existent by 1915, I assume that the industry reaches a stationary equilibrium (i.e., industry demise) at that point and solve for the nonstationary equilibrium backward from there with a terminal value function of zero.

I now define the dynamic competitive equilibrium for a common-pool resource industry.

²⁵The wreck hazard p_t is calibrated from the firm-year panel as the share of operating firms that lost all of their vessels to a wreck in year t and did not operate again.

²⁶Note that firms form beliefs about the future whale stock $\tilde{W}_{t+1}$ (instead of W_{t+1}), subject to uncertainty.

²⁷Equations (12) and (13) normalize the operating fixed cost to zero. This normalization is necessary because, in a dynamic setting, the entry cost, exit value, and operating fixed cost cannot be jointly identified (Aguirregabiria and Suzuki 2014). As a result, estimates of entry cost and exit value incorporate the operating fixed cost.

Given an initial firm distribution f_1 and resource stock W_1 , an equilibrium is a sequence of decision rules $\{\psi_t\}_{\forall t}$, a price path $\{P_t\}_{\forall t}$, a stock path $\{W_t\}_{\forall t}$, and an aggregate trajectory $\{K_t, Q_t\}_{\forall t}$ such that:

1. Incumbents solve (12) and potential entrants solve (13), taking the price and the industry aggregates as given.
2. In every year, the market price is given by (5), total vessel capacity is given by (7), and total whale harvest is given by (8).
3. The firm distribution evolves according to (10).
4. The whale stock evolves according to (1).

Existence follows from the oblivious-equilibrium arguments of Weintraub, Benkard, and Van Roy (2008) and their nonstationary extension in Benkard, Jeziorski, and Weintraub (2024). In this dynamic competitive equilibrium, each firm solves a single-agent problem; the strictly convex adjustment cost pins down a unique capacity choice, and the continuous shocks smooth the exit and adjustment margins.

4. Estimation

The estimation proceeds in two stages. In the first stage, I estimate the whale harvesting (production) function and the whale demand curve (Section 4.1). These estimates provide period payoffs for all possible firm states and years. In the second stage, I embed these payoffs into the dynamic model and estimate the dynamic parameters (Section 4.2). Table 2 summarizes the structural parameters that are estimated and calibrated. Panel A reports the primitives from the harvesting function and demand curve. Panel B lists the dynamic parameters. Panel C includes calibrated values.

4.1. Period payoffs

Whale harvesting function. I parameterize equation (4) as a Cobb–Douglas production function in logarithmic form:

$$(14) \quad q_{it} = \beta_0 + \beta_k k_{it} + \beta_l l_{it} + \beta^K k_t + \beta^W w_t + \beta^t t + \underbrace{\mu_{it} + \varepsilon_{it}}_{\omega_{it}},$$

where q_{it} represents the log of the number of harvested whales (sperm whales + baleen whales), k_{it} is the log of vessel capacity, l_{it} is the log of crew size, k_t is the log of aggregate vessel capacity, w_t is the log of whale stock, and t is the time trend. The firm-level (log) productivity, ω_{it} , is divided into persistent productivity μ_{it} and a transitory shock ε_{it} .

TABLE 2. List of parameters and empirical approach

Parameters	Notation	Empirical approach
Panel A. Period payoffs		
Harvesting function	$\beta_0, \beta_k, \beta^K, \beta^W, \beta^t, \lambda$	DP-GMM (Section 4.1)
Demand curve	$\alpha_0, \alpha_p, \alpha^{\text{gdp}}, \alpha^t, \alpha^{\text{pet}}$	OLS, IV (Section 4.1)
Panel B. Dynamic costs and values		
Mean of entry costs	$\bar{\kappa}$	NFXP-GMM (Section 4.2)
Mean of exit values	$\bar{\zeta}$	NFXP-GMM (Section 4.2)
Scale of entry costs and exit values	σ_L	NFXP-GMM (Section 4.2)
SD of stochastic marginal adjustment costs	σ	NFXP-GMM (Section 4.2)
Deterministic adjustment costs	$\gamma_0^+, \gamma_1^+, \gamma_2^+, \gamma_0^-, \gamma_1^-, \gamma_2^-$	NFXP-GMM (Section 4.2)
Panel C. Transitions		
Annual discount factor	$\rho = 0.95$	Calibrated
Vessel capacity depreciation rate	$\delta = 0.05$	Calibrated
Whale stock dynamics	$z = 1.4, r = 0.011$	Calibrated
Belief dispersion over the whale stock	$\sigma_W = 0.10$	Calibrated
Destruction hazard	p_t	Calibrated

Note: This table summarizes key parameters of the model. Parameters in panels A and B are estimated, whereas those in Panel C are calibrated. In panel A, DP-GMM refers to dynamic panel estimation using the generalized method of moments (GMM). In panel B, NFXP-GMM denotes nested fixed-point estimation by the GMM. Parameters related to whale population dynamics are drawn from previous literature (Baker and Clapham 2004; Whitehead 2002).

Following the dynamic panel approach by Blundell and Bond (2000), persistent productivity μ_{it} evolves as an AR(1) with serial correlation λ :

$$(15) \quad \mu_{it} = \lambda \mu_{it-1} + \xi_{it},$$

where ξ_{it} is an independent and identically distributed (i.i.d.) shock.²⁸

Firms choose vessel capacity in year $t - 1$ (as in Eq. 2) and crew size between $t - 1$ and t (as in Akerberg, Caves, and Frazer 2015), both prior to the year t productivity shock ξ_{it} . Combined with the AR(1) process (Eq. 15), this timing yields the exclusion restriction

²⁸Imposing this AR(1) structure rules out the richer productivity transition in equation (3) and may not fully address selection bias from endogenous exit. Fully resolving the selection issue would require estimating the production function jointly with firms' entry, exit, and capacity adjustment decisions, which could be an extension I leave for future work. A common alternative is an approach that inverts intermediate-input demand to recover latent productivity (Levinsohn and Petrin 2003; Akerberg, Caves, and Frazer 2015). This approach requires an intermediate input whose demand is monotonically increasing in productivity. Because the data for the American whaling industry lack such an input, this approach is not feasible here. Instead, in Online Appendix Table E4, I test the dynamic estimates by increasing $\hat{\beta}_k$, because endogenous exit biases $\hat{\beta}_k$ downward. The results are not sensitive to this change.

TABLE 3. Estimates of the whale harvesting function

	(1)	(2)	(3)	(4)
Vessel capacity: β_k	1.021 (0.029)	0.867 (0.029)	1.057 (0.009)	1.048 (0.008)
Crew size: β_l	-0.003 (0.026)	0.239 (0.032)		
Persistence: λ	0.655 (0.101)	0.269 (0.114)	0.373 (0.116)	0.300 (0.087)
Whale stock: β^W		6.351 (0.311)	5.949 (0.337)	5.101 (0.270)
Aggregate vessel capacity: β^K		-0.042 (0.016)	-0.026 (0.017)	-0.057 (0.010)
Time trend: β^t		0.016 (0.002)	0.016 (0.002)	0.013 (0.001)
Returns to scale: $\beta_k + \beta_l$	1.017	1.106	1.057	1.048
Number of firms	440	440	440	874
Observations	5,022	5,022	5,022	8,963

Notes: Columns (1)–(3) use the subsample with crew data; column (4) employs the full sample. Identification relies on the following instruments: contemporaneous and lagged vessel capacity, lagged crew size, contemporaneous aggregate capacity, whale-stock abundance, and a linear time trend. Parameters are estimated by generalized method of moments (GMM). Robust GMM standard errors, based on the estimator’s asymptotic variance–covariance matrix, appear in parentheses.

identifying $\beta = (\beta_0, \beta_l, \beta_k, \beta^K, \beta^W, \beta^t, \lambda)$. Specifically, the year- t productivity shock ξ_{it} is orthogonal to $z_{it} = (1, l_{it-1}, k_{it}, k_{it-1}, k_t, w_t, t)$. The corresponding moment conditions are

$$\mathbb{E} \left[\overline{\xi_{it} + (\varepsilon_{it} - \lambda \varepsilon_{it-1})} (\beta) \otimes z'_{it} \right] = 0.$$

I estimate the parameters by GMM.

Table 3 reports estimates of the whale harvesting function. Columns (1)–(3) use the subsample that contains crew data, while column (4) uses the full sample and serves as the baseline for the dynamic analysis. Column (1) only includes vessel capacity and crew size; column (2) adds aggregate time-series variables. Column (3) excludes crew size from the same subsample to compare with the full-sample specification in column (4).

The estimated results show that vessel capacity alone, without crew size, explains most of the variation in production scale. This pattern is intuitive because nineteenth-century American whaling was capital-intensive, and crew size was roughly proportional to vessel capacity. Accordingly, in the analyses that follow, I assume that harvesting combines vessel

capacity and crew in fixed proportions, reflecting a Leontief technology.²⁹

Based on the results of column (4), a back-of-the-envelope calculation suggests strong production externalities. First, the coefficient for whale stock (β^W) captures the stock externality. Holding other factors constant, a 1 percent increase in whale stock raises an individual firm's harvest by 5.1 percent. Because whale stock fell by 15 percent between 1820 and 1850, the implied harvest reduction is 56.1 percent. Second, the coefficient for aggregate capacity (β^K) captures the congestion externality. A 1 percent increase in aggregate capacity reduces an individual firm's harvest by 0.057 percent. Between 1820 and 1850, aggregate capacity rose from 37,315 to 178,096 tons (an increase of 377 percent). Applying the elasticity estimate implies an 8.6 percent decline in an individual firm's harvest due to congestion.

Demand for whales. The whaling market features a demand curve in the form of equation (5), inverted and parameterized as:

$$(16) \quad \ln Q_t = \alpha_0 + \alpha_p \ln P_t + \alpha^{\text{gdp}} \ln \text{GDP}_t + \alpha^t t + \alpha^{\text{pet}} \ln \text{Pet}_t + \eta_t,$$

where P_t is the whale price, GDP_t denotes U.S. real GDP per capita, t is a time trend, and Pet_t , included for $t \geq 1859$, represents the price of petroleum. η_t is an i.i.d. demand shock. I estimate this specification separately before and after 1859, as demand likely shifted after the discovery of petroleum (see Section 2.1 for details).

The whale price may be endogenous because it can co-move with the demand shock η_t . Such shocks shift the demand curve and the equilibrium price, biasing the estimated price elasticity. I address this concern with two supply-side cost shifters as instruments for price. They move supply costs and hence prices but are orthogonal to η_t conditional on controls.

First, I use whale stocks lagged five years. Because whales are slow-growing, past abundance strongly predicts current stock levels, which affect harvest costs. A five-year lag also exceeds the typical voyage length of two to three years, reducing simultaneity with current harvest outcomes. Current demand for whale products cannot affect whale abundance five years earlier. Any reverse channel would require persistent unobserved shocks that link past harvesting effort to current demand. Macroeconomic controls (GDP per capita and a time trend) absorb the main aggregate demand drivers. Any remaining correlation would have to come from unobserved demand shocks with unusually long

²⁹I therefore model crew size as $L(K_{it}) = cK_{it}$, where c denotes crew per tonnage. The data support this assumption: vessel tonnage and crew size are tightly linked. The correlation coefficient between capacity K_{it} and crew size L_{it} is 0.89; between their logs k_{it} and l_{it} it is 0.88.

TABLE 4. Estimates of the whale demand curve

	OLS		IV	
	(1)	(2)	(3)	(4)
Whale price: α_p	-1.980 (0.960)	-0.230 (0.070)	-4.390 (0.787)	-4.390
US GDP per capita	-4.380 (4.533)	0.458 (0.257)	3.081 (2.555)	2.044 (1.768)
Time trend	0.168 (0.035)	-0.047 (0.005)	0.137 (0.025)	-0.109 (0.034)
Petroleum price		0.057 (0.034)		-0.355 (0.356)
Observations	55	51	51	51
Sample period	1804-1858	1859-1909	1804-1858	1859-1909
First-stage F -statistic			38.407	
Overid. test (p -value)			0.24	
Adjusted R-squared	0.822	0.956	0.769	0.501

Notes: Columns (1) and (2) present OLS results for the pre- and post-petroleum periods, respectively. Column (3) reports instrumental-variables estimates for the pre-petroleum sample, instrumenting the whale price with two supply-side shifters: the whale stock lagged five years and long-term New England bond yields. Column (4) presents post-petroleum estimates, fixing the price coefficient, $\hat{\alpha}_p$, at the elasticity from column (3). The first-stage F -statistic is the robust Wald statistic for the excluded instruments; the overidentification test reports the p -value of Hansen's J statistic for their joint validity, where a large p -value fails to reject. Standard errors are reported in parentheses (Newey-West for the OLS columns, heteroskedasticity-robust for the GMM column).

persistence that both influenced harvest effort five years earlier and predicted current U.S. demand, which seems implausible.

Second, I exploit variation in long-term bond yields in New England.³⁰ Whaling was highly capital-intensive; firms relied on external finance to purchase ships, outfit crews, and finance long voyages (see Section 2.2). Higher bond yields increase borrowing costs and therefore raise the cost of supply. Whale products, by contrast, were sold nationally and internationally, so the marginal buyer was a household or mill far from New England. After controlling for aggregate U.S. demand conditions (GDP per capita and a time trend), regional long-term borrowing costs should be orthogonal to shocks in whale-product demand. Local credit conditions largely reflected banking cycles, lending practices, and regional capital scarcity rather than demand for whale products (Lamoreaux 1994).

I restrict the IV strategy to the pre-petroleum era (before 1859) for two reasons. First, whale stock exhibits little variation after 1859, undermining its validity as an instrument

³⁰The data are constructed by Homer and Sylla (1996).

(Panel B of Figure 1). Between 1859 and 1909, whale stocks remained nearly constant (1.16–1.22 million whales), in contrast to the sharp decline from 1.48 to 1.19 million by 1858. Second, by mid-century the whaling center shifted from New Bedford (and New England more broadly) to San Francisco, making New England bond yields no longer a credible cost shifter. Accordingly, for the post-petroleum period, I calibrate the price coefficient α_p using the elasticity estimated from the pre-petroleum period. I also consider a reasonable range of alternative price elasticities, re-estimate the dynamic model, and recompute the counterfactuals (see Online Appendix Table E3). The main welfare and policy implications are not sensitive to these assumptions.

Table 4 reports estimates of the whale demand curve. Columns (1) and (2) present OLS estimates for the pre- and post-petroleum eras, with price elasticities of -1.980 and -0.230 , respectively. Column (3) shows IV estimates for the pre-petroleum sample; the elasticity rises in absolute magnitude to -4.390 , consistent with Kaiser (2013). For the post-petroleum period, I fix the price coefficient α_p at -4.390 and estimate the remaining parameters via OLS. The results in columns (3) and (4) form the basis for subsequent dynamic analyses. Online Appendix Table E3 repeats the dynamic estimation and the counterfactuals of Sections 4.2 and 5 with the post-petroleum price coefficient fixed at -2.0 and at -5.0 .

4.2. Dynamic costs and values of whaling

I estimate the dynamic parameters by nested fixed-point (NFXP) generalized method of moments (GMM). Online Appendix D discusses the choice of this full-solution method. For each candidate parameter vector, I solve the equilibrium, which gives the probabilities of exit, of each capacity level, and of entry at every state and year.

The main computational challenge comes from continuous capacity adjustment. Computing optimal decision rules for dynamic actions can be highly sensitive to how the continuous state (and action) space is discretized. To address this issue, I adopt the approach of Gowrisankaran and Schmidt-Dengler (2025), which introduces random shocks to marginal adjustment costs and yields choice probabilities that are robust to the discretization. Under this approach, a capacity level that no draw of the adjustment-cost shock makes optimal (dominated action) has probability zero, which leaves the log likelihood undefined, so I estimate by GMM, as in Gowrisankaran, Langer, and Reguant (2026). Online Appendix A.1 derives the choice probabilities and defines the moments.

The moments compare firms' observed decisions with the model's predictions at each firm's state and year. On the exit decision, the moment is the difference between the data and the model in the exit rate. On the capacity-level choice, among staying firms, the moments match the shares of firms that invest and that divest and the amounts invested

TABLE 5. Estimates of whaling industry dynamics

	Unit	1815-29	1830-44	1845-59	1860-74	1875-89	1890-1905
Panel A. GMM estimates							
Mean of entry costs: $\bar{\kappa}$	\$000	373.22 (60.84)	410.75 (43.63)	892.66 (70.12)	1156.62 (148.77)	1209.59 (295.06)	910.44 (190.98)
Mean of exit scrap values: $\bar{\zeta}$	\$000	275.67 (53.16)	302.29 (33.92)	655.99 (55.86)	775.70 (119.96)	966.24 (251.34)	659.30 (137.20)
Scale parameter: σ_L	\$000	40.57 (7.57)	45.84 (5.44)	69.75 (7.47)	114.42 (15.94)	89.03 (22.50)	78.04 (29.42)
Investment fixed: γ_0^+	\$000	18.47 (5.84)	4.38 (1.32)	5.04 (1.26)	0.00 (4.63)	0.00 (7.85)	2.10 (14.66)
Investment linear: γ_1^+	\$/ton	300.02 (60.99)	285.95 (33.20)	363.56 (27.88)	613.56 (99.90)	911.68 (306.39)	980.52 (239.35)
Investment convex: γ_2^+	\$/ton ²	0.57 (0.15)	0.24 (0.05)	0.28 (0.04)	0.45 (0.12)	0.71 (0.33)	1.08 (0.46)
Divestment fixed: γ_0^-	\$000	46.74 (14.59)	41.54 (8.00)	40.04 (5.50)	64.21 (13.60)	119.82 (60.25)	124.02 (66.34)
Divestment linear: γ_1^-	\$/ton	61.95 (69.98)	26.73 (41.54)	111.50 (26.32)	164.02 (38.10)	76.02 (176.06)	0.00 (270.02)
Divestment convex: γ_2^-	\$/ton ²	0.85 (0.32)	0.28 (0.06)	0.26 (0.04)	0.54 (0.12)	1.17 (0.61)	1.26 (0.73)
SD of marginal adj. costs: σ	\$/ton	345.30 (93.55)	231.17 (46.73)	227.45 (30.16)	436.69 (93.00)	805.21 (383.94)	816.21 (337.09)
Number of observations		1015	2950	3246	1626	709	342
Panel B. Realized magnitudes							
Entry cost	\$000	302	336	724	870	1027	719
Exit value	\$000	402	453	877	1114	1230	882
Investment of 300 tons	\$000	14	26	44	32	4	11
Divestment of 300 tons	\$000	70	52	68	97	118	118

Note: Each column presents results for the corresponding period. Panel A summarizes two-step GMM estimates. Standard errors, shown in parentheses, are computed from the standard GMM formulas. Panel B reports the average cost or value realized by the firms that took each action, over 200 simulations of the estimated model. Investment and divestment are for a change of 300 tons, close to the median adjustment size in Table 1, and divestment is the amount the firm receives. All dollar values are adjusted to 1880 prices.

and divested. Further moments match the squares of the two amounts and the shares of firms that move more than 300 tons up or down. A last moment matches the dispersion of net investment around the model's conditional mean. On the entry margin, the moment is the difference between the number of entrants in a year and the number the model predicts. I interact the exit moments and the shares of firms that invest and divest with the firm's capacity and productivity, the amounts invested and divested with capacity, and the exit and entry moments with the year. Online Appendix A.1 defines the 21 moments and discusses identification.

I estimate the model in six phases: 1815–1829, 1830–1844, 1845–1859, 1860–1874, 1875–1889, and 1890–1905. I divide the sample this way because the industry faced structural breaks that plausibly shifted firms’ beliefs (see Section 2.1). The arrival of petroleum and the Civil War make 1860 a natural breakpoint, and the mid-1840s shocks (the California gold rush, longer voyage durations, etc.) motivate a separate phase beginning in 1845, because firms did not anticipate them. I divide the remaining phases into 15-year intervals for consistency. Finally, because the War of 1812 (1812–1815) severely disrupted the industry and left little firm-level data, the estimation sample for dynamic parameters begins in 1815.

Table 5 reports the dynamic parameter estimates. Panel A shows that the estimates are economically plausible and precisely estimated. Mean entry costs range from \$373,217 to \$1,209,593 across the six phases, low during the industry’s expansion through 1844 and high from 1845 on. Mean exit scrap values are lower in every phase, ranging from \$275,665 to \$966,241. Low entry costs and low exit values during the expansion match the observed turnover, with heavy entry and limited exit during the industry’s growth. Adjustment cost estimates indicate sizable frictions in changing capacity. The linear cost terms imply that investment costs rise with scale, ranging from \$285.95 to \$980.52 per ton, while divestment values range from \$0.00 to \$164.02 per ton.³¹ Convex cost parameters range from \$0.24 to \$1.08 per ton squared for investment and from \$0.26 to \$1.26 for divestment. Finally, the standard deviation of marginal adjustment-cost shocks, ranging from \$227.45 to \$816.21 per ton, indicates substantial heterogeneity in adjustment intensity.

Panel B of Table 5 reports the costs and values that firms realized, averaged over the firms that took each action in 200 simulations of the estimated model. Firms enter on low cost draws, leave on high exit values, and adjust on favorable shocks, so the realized magnitudes differ from the (unconditional) means in Panel A. Adding 300 tons of capacity cost \$44,000 in 1845–1859. According to the historical record, buying a 300-ton ship and outfitting a 39- to 46-month voyage cost \$41,000 to \$46,000 in the same years (Davis, Gallman, and Gleiter 2007). Divesting 300 tons returned \$52,000 to \$118,000 across the phases, implying that firms divested only when they found it profitable to do so. Realized entry costs and exit values are an order of magnitude larger, \$302,000 to \$1,027,000 and \$402,000 to \$1,230,000, because the entry cost includes the initial business setup cost and the various barriers to becoming a whaling firm, and the exit value captures the firm’s next-best use of the capital (opportunity cost of staying in the whaling industry).

³¹When firms divest, a positive γ_1^- implies a positive linear “value.” Total adjustment costs should be assessed jointly with fixed and convex costs; see Panel B of Table 5.

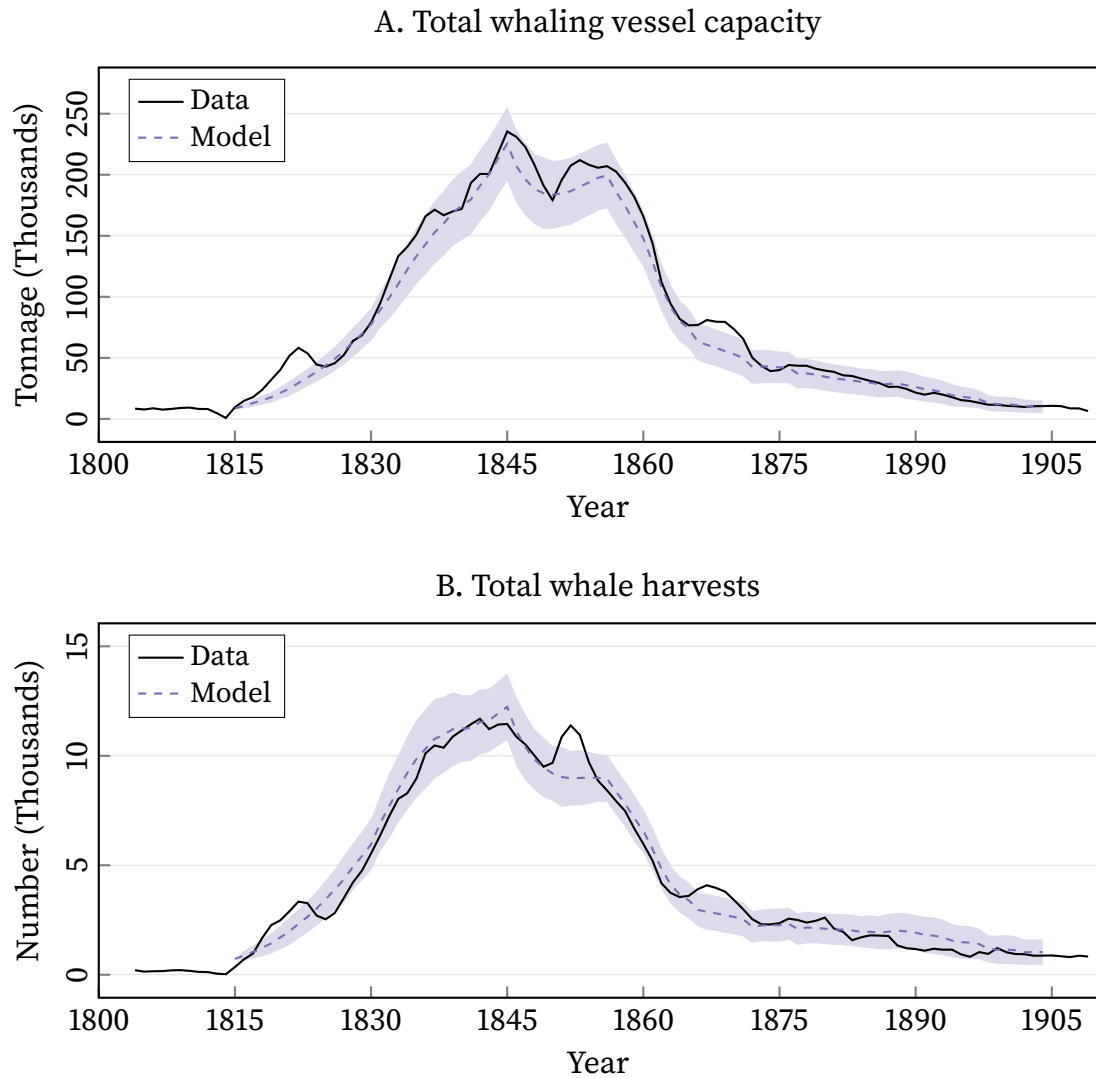


FIGURE 3. Fit of the estimated model

Note: Panel A shows total whaling-vessel capacity and panel B shows total whale harvest. The black solid line indicates the actual data, and the purple dashed line shows the model-predicted paths. Model outcomes are averaged over 200 simulations of the estimated model, with shaded bands denoting 95 percent confidence intervals.

I evaluate model fit by comparing observed behavior in the data with predictions from the estimated model. Figure 3 compares the data (solid line) with model-implied paths (dashed line) for total whaling-vessel capacity and total whale harvest. The model paths are averages across 200 simulations under the estimated parameters, with shaded bands indicating 95 percent confidence intervals. The figure shows that the model reproduces the industry's rise and fall, including the expansion through the mid-1840s and the subsequent long decline.

5. Counterfactuals

This section uses the estimated model to measure the social cost of the externality over the long run. Section 5.1 states the planner's problem, recovers the shadow value of a whale, and draws a simple policy implication. Section 5.2 reports the welfare loss under open access relative to the social planner. Section 5.3 asks how the demand for whale products and the industry's technology affect the size of that loss.

5.1. The planner's problem and the shadow value of a whale

I define the social welfare function in year t as the sum of consumer and producer surplus:

$$(17) \quad \mathbb{W}_t = \underbrace{\int_0^{Q_t} \mathcal{P}_t(\varphi) d\varphi - \mathcal{P}_t(Q_t)Q_t}_{\text{consumer surplus}} + \underbrace{\sum_{i \in \mathcal{J}_t} \left\{ \Pi_t(s_{it}) + \mathbf{1}_{\{a_{it}=0\}} \zeta_{it} - \mathbf{1}_{\{a_{it}>0\}} [C(I_{it}) + I_{it}\epsilon_{it}] \right\} - \sum_{i \in \mathcal{J}_t^{\text{pe}}} \mathbf{1}_{\{a_{it}^{\text{pe}}=1\}} \kappa_{it}}_{\text{producer surplus}}$$

A social planner chooses incumbents' exit and capacity adjustments $\{a_{it}\}_{\forall i,t}$ and potential entrants' entry $\{a_{it}^{\text{pe}}\}_{\forall i,t}$ to maximize the discounted sum of social welfare (17) subject to the whale stock's law of motion (1). The Lagrangian of this maximization problem is

$$(18) \quad \mathbb{E} \left[\sum_t \rho^{t-1} \left\{ \mathbb{W}_t + \rho \Lambda_{t+1}^W \left((W_t + rW_t[1 - (W_t/W_1)^z] - Q_t) \Phi_{t+1}^W - W_{t+1} \right) \right\} \right],$$

where Λ_{t+1}^W is the multiplier on the whale-stock constraint. The expectation is taken with respect to the random cost shocks $(\zeta_{it}, \epsilon_{it}, \kappa_{it})$, the productivity draws (Ω_{it}) , and the stock shock ϕ_{t+1}^W .

The shadow value of a whale. The multiplier Λ_{t+1}^W measures the discounted value of every future harvest that one more whale in the stock would have yielded.³² That is, it is the shadow value of a whale in the year- $(t+1)$ stock, measured in year- $(t+1)$ dollars. Importantly, what that whale is worth depends on the number of firms and vessels operating in later years, which entry, exit, and capacity adjustment determine.

Figure 4 shows the implied shadow value. In panel A, the solid line is the shadow value

³²Online Appendix A.2 derives the shadow value from the planner's first-order condition.

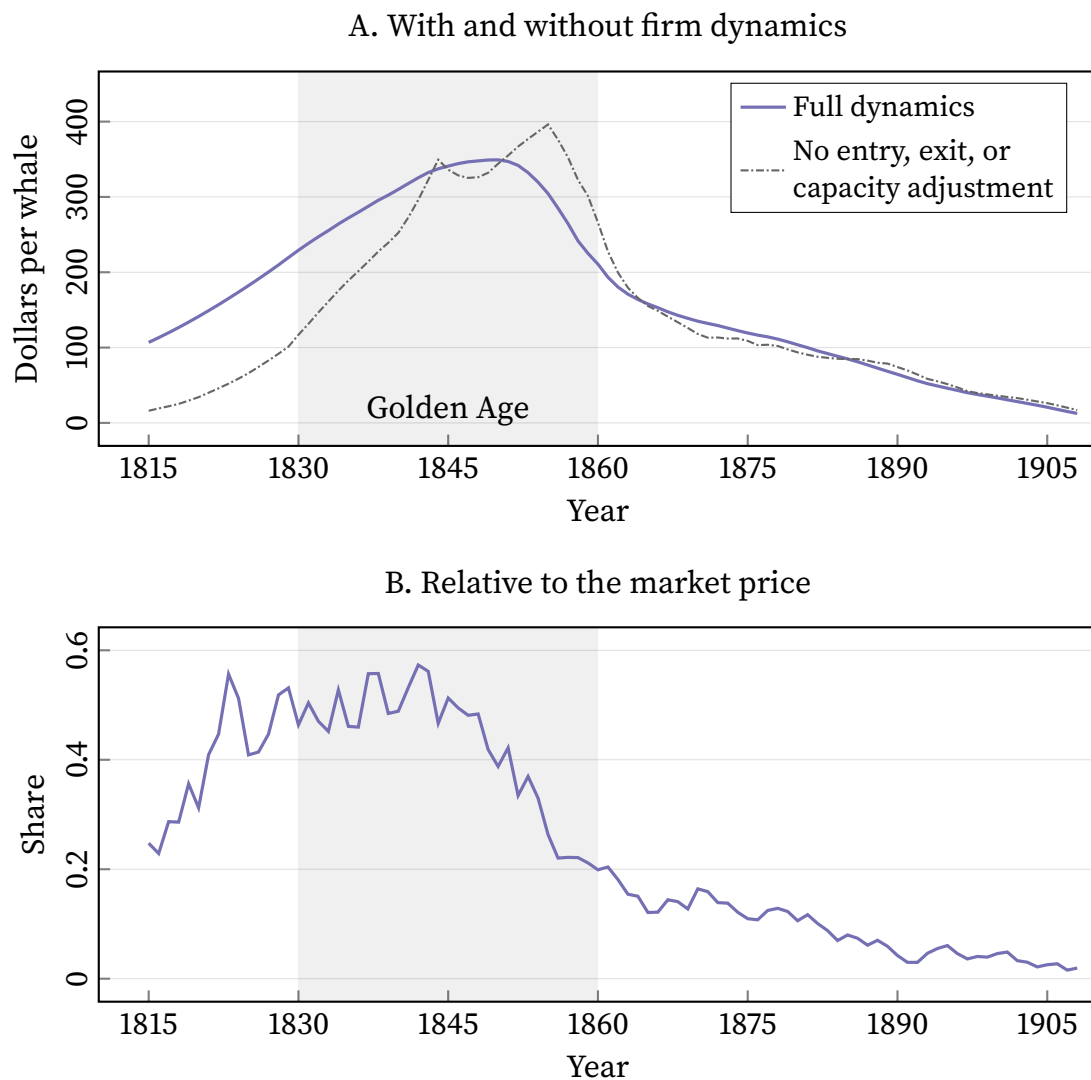


FIGURE 4. The shadow value of a whale

Note: In panel A, the solid line is the whale shadow value in the equilibrium with entry, exit, and capacity adjustment. The dash-dotted line removes all three dynamics. Panel B divides the solid line of panel A by the observed market price of a whale in the same year. All dollar values are adjusted to 1880 prices. The shaded gray period marks the Golden Age (1830–1860).

under the full dynamic model with entry, exit, and capacity adjustment; the dash-dotted line removes all three, so only the whale stock evolves. A whale harvested in 1815 has a true social value of \$107 (adjusted to 1880 prices) with firm dynamics and \$16 without them. One more whale harvested lowers the catch of the vessels at sea in later years. The difference between the two lines is therefore the part of the external cost that comes from how many firms and vessels operate, which is determined by firm dynamics. Over the early boom (1815–1830), those dynamics account for 61 percent of the external cost of the whales harvested. In the 1850s the ordering reverses. By 1855 the full shadow value is

TABLE 6. A tax on the catch set at the measured shadow value of a whale

	Tax (\$ per whale)		Social welfare
	First	Highest	
Laissez-faire	0	0	30.3
<i>Legislated once, in force from 1831</i>			
Without entry, exit, and investment	72	72	34.0 [+3.7]
With entry, exit, and investment	185	185	38.1 [+7.7]
<i>Revised every year</i>			
Without entry, exit, and investment	16	387	40.5 [+10.2]
With entry, exit, and investment	107	320	43.5 [+13.2]

Note: “Legislated once” means that the value is measured over 1815–1830 and the tax is in force from 1831 onward. “Revised every year” means that each year from 1816 to 1905 the regulator posts the value measured over the year just ended, and firms treat the posted rate as permanent until it changes. All dollar values are adjusted to 1880 prices. Laissez-faire is the estimated equilibrium. Social welfare is the sum of consumer surplus, producer surplus, and tax revenue over 1815–1915, in millions of dollars, discounted to 1815 at $\rho = 0.95$ and averaged over 300 simulations. The welfare gain over laissez-faire is in square brackets.

\$304 while the dash-dotted line is \$396, because the industry that would take that whale is by then shrinking rather than growing. In sum, overlooking entry, exit, and capacity adjustment understates the shadow value of whales in the early boom and overstates it before the industry declines.

Panel B divides the whale shadow value (with firm dynamics) by the market price of a whale in the same year. The share rose from 25 percent in 1815 to near one-half by the 1830s and stayed there through 1850, peaking at 57 percent in the early 1840s. It fell to 21 percent by the late 1850s, as the market price rose and the shadow value began to fall. The share was 11 percent by 1875 and 4 percent by 1890. After the petroleum discovery and the Civil War, the industry shrank, so one more whale in the stock would have raised later harvests by less.

The value measured without firm dynamics is the one a regulator would typically compute in practice. Valuing a resource stock means projecting what the industry will take from it in future years, and the international standard for natural-capital accounting projects extraction forward at a constant or the most recent rate, which holds the industry structure fixed. For instance, consider *System of Environmental-Economic Accounting 2012: Central Framework* (United Nations 2014), Chapter V. The framework values a resource stock as the net present value of the future resource rents it is expected to yield, and for projecting future production it suggests “a constant rate of extraction or the most recent quantity of extraction.”

In Table 6, I illustrate the consequences of this approach with a tax on each whale

TABLE 7. The welfare cost of open access

	Laissez-faire	First best	Difference
Social welfare	30.32	44.68	14.36
Share of first best (%)			[32.1]
<i>By era</i>			
Pre-1859	14.70	39.28	24.58
Post-1859	15.62	5.40	-10.22
<i>By welfare component</i>			
Consumer surplus	13.66	7.36	-6.30
Producer surplus	16.65	37.32	20.67

Note: All entries are in millions of 1880 dollars, discounted to 1815 at $\rho = 0.95$ and averaged over 300 simulations. Laissez-faire is the estimated equilibrium; the first best solves problem (18). The era rows split each allocation's welfare at the petroleum discovery, and the component rows split it into consumer and producer surplus.

caught, set at the shadow value measured *with* and *without* firm dynamics. I consider two designs. Under the first (and simplest), a regulator measures the shadow value over 1815–1830, before the over-harvesting of the Golden Age. It imposes the tax from 1831 onward, based on the average shadow value over 1815–1830. Under the second, the regulator re-measures the shadow value every year based on the industry it observes and imposes the new rate for the following year. This design is more demanding than most tax regimes in the real world, since the regulator must revalue the stock and revise the rate every year. Hence, in practice, a feasible policy design would be somewhere between the first and the second. In the first design, the tax measured with entry, exit, and capacity adjustment over 1815–1830 is \$185 per whale, and the tax measured without them is \$72. It delivers only 47 percent of the welfare gain that the correctly measured tax delivers. In the second design, the rate measured without firm dynamics starts at \$16 against \$107 and stays below the correct rate in every year through 1852. It then overshoots, peaking at \$387 against \$320. In this case, the tax measured without firm dynamics forgoes 22 percent of the welfare gain that the correct one delivers. These results suggest that a policy ignoring firm dynamics would deliver only 47 (legislated once) to 78 (revised every year) percent of the welfare gain available under full industry dynamics.

5.2. The cost of open access

Table 7 summarizes welfare under open access (laissez-faire) and under the social planner (first best). Laissez-faire delivers \$30.32 million and the first best \$44.68 million, so open access costs \$14.36 million, 32 percent of first-best welfare. The gain is concentrated in

the industry's growth period. Before 1859 the first best is worth \$24.58 million more than laissez-faire, because most of the external cost arises in the early years. By component, consumer surplus falls from \$13.66 million to \$7.36 million, because the planner harvests fewer whales and the price of a whale is higher. Producer surplus rises from \$16.65 million to \$37.32 million.

Figure 5 shows the mechanism of the pre-1859 gain and the rise in producer surplus. Panel A plots total vessel capacity under laissez-faire and under the first best. Laissez-faire vessel capacity peaks at 224 thousand tons in the mid-1840s, 4 times the first-best peak of 57 thousand tons. Panel B shows the impact on the whale stock. Under laissez-faire the stock falls to 1.16 million whales at its lowest, after the Golden Age. Under the first best it never falls below 1.36 million. Panel C shows the efficiency consequence. It plots the number of whales harvested per thousand tons of vessel capacity. Under laissez-faire (which closely follows the black data line), harvesting efficiency falls from 83.4 whales per thousand tons in 1815–1819 to 45.1 in 1855–1859, a decline of 46 percent. The first best moves the other way. It rises by 34 percent over the same window. By the late 1850s a ton of first-best capacity harvests 2.5 times what a ton under laissez-faire harvests. Under open access, every additional ton draws down the whale stock and crowds the hunting grounds, so each ton harvests far less than it would under the first best.

5.3. Whaling's fortunes and externalities

The welfare loss in Table 7 is based on the demand and technology paths the industry actually faced. Demand for sperm oil and whale oil grew with the American economy until the petroleum discovery of 1859 reduced it, and harvesting technology gradually improved throughout the century. In this subsection, I examine how each of these long-run changes affected the welfare cost of open access.

The petroleum discovery. Petroleum was discovered at Titusville, Pennsylvania, in 1859, and demand for whale products fell away over the following decade. In Table 8, I ask what open access would have cost had the substitute arrived later, in 1875 or 1890, or never. Ex ante the answer is not obvious. A longer period of high demand would have made the industry more profitable, but it would also have drawn in more firms and vessels, which would have worsened the externality.

Laissez-faire welfare falls with each later arrival date, from \$30.32 million to \$18.10 million. In contrast, first-best welfare rises with each later arrival date, from \$44.68 million to \$48.97 million. The commons loss therefore rises, from \$14.36 million to \$23.14 million and \$28.42 million, and to \$30.88 million with no petroleum discovery at all. As a share

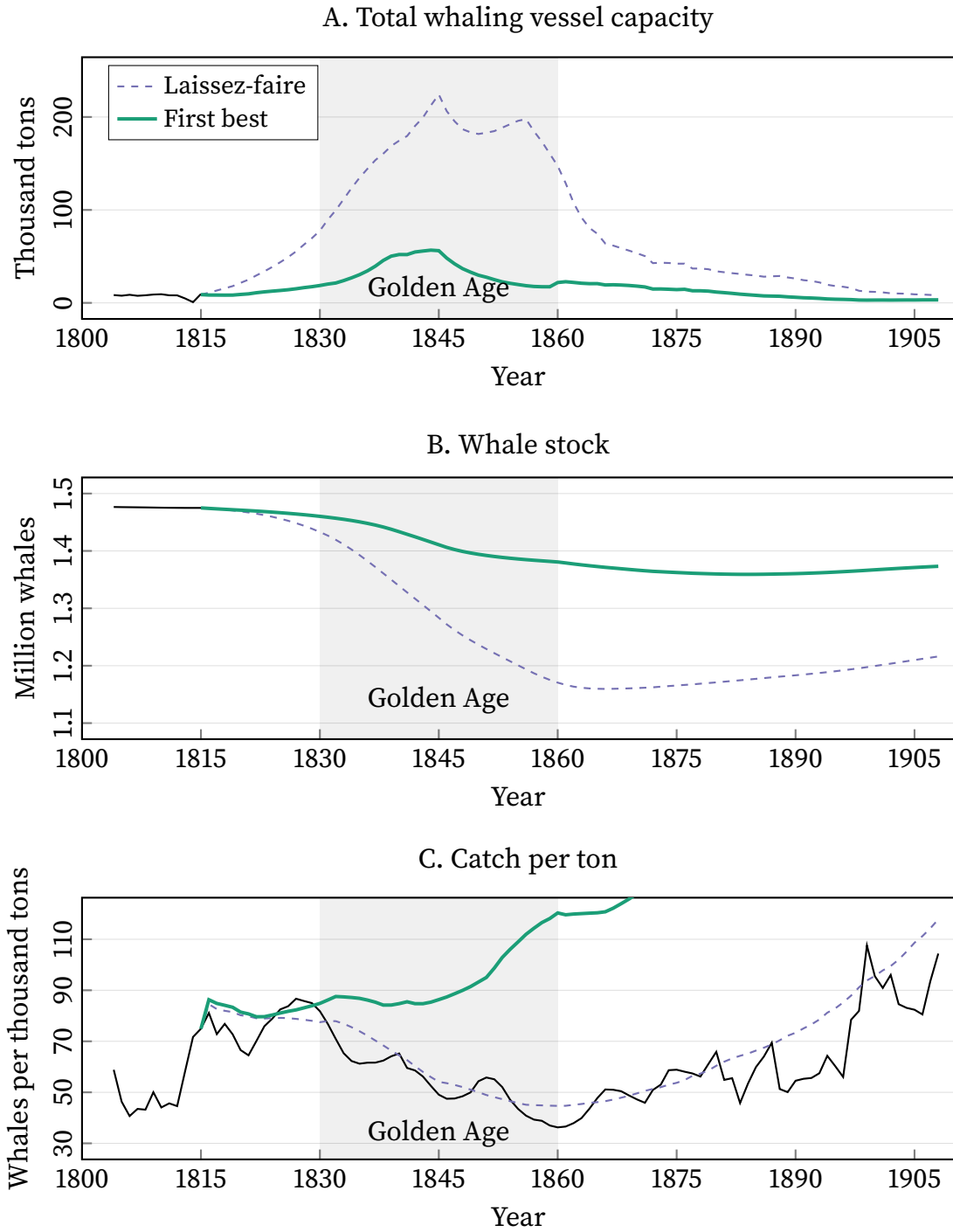


FIGURE 5. Vessel capacity and harvesting efficiency

Note: Panel A shows total whaling vessel capacity in thousand tons. Panel B shows the whale stock. Panel C shows whales harvested per thousand tons of vessel capacity, and the black line is the observed series over the whole sample. In all panels the purple dashed line is the laissez-faire equilibrium and the green solid line is the first best. The shaded gray period marks the Golden Age (1830–1860).

TABLE 8. The cost of open access by petroleum arrival date

	Petroleum arrival year			
	1859	1875	1890	Never
<i>Panel A. Social welfare</i>				
Laissez-faire (\$M)	30.32	24.14	20.03	18.10
First best (\$M)	44.68	47.28	48.45	48.97
Difference (\$M)	14.36	23.14	28.42	30.88
Share of first best (%)	[32.1]	[48.9]	[58.7]	[63.0]
<i>Panel B. Laissez-faire outcomes</i>				
Lowest stock level (Million whales)	1.16	1.13	1.09	1.05
Peak capacity (Thousand tons)	224.2	224.2	224.2	224.2

Note: Each column re-solves laissez-faire and the first best with the petroleum discovery moved to the stated year. In panel A, social welfare is in millions of 1880 dollars, discounted to 1815 at $\rho = 0.95$ and averaged over 300 simulations. Panel B reports the lowest stock level and the peak capacity from the laissez-faire equilibrium.

of first-best welfare, it rises from 32 to 63 percent. The arrival of petroleum in 1859 thus alleviated the cost of open access to less than half of what it would have been. Under open access, the whale stock's lowest level falls to 1.05 million whales with no arrival, against the 1.16 million actually reached. The added external cost of the extra entry and investment outweighs the direct gain from longer-lasting demand. Therefore, higher demand turns into welfare gains only when the externality is properly addressed.

Technology growth. The harvesting (production) function estimates imply overall productivity growth of about 1.3 percent per year (see Table 3). The growth reflects better gear, larger and faster vessels, and accumulating knowledge of the hunting grounds, from the toggle iron of 1848 to bomb lances in the 1850s and steam-powered whalers by the 1880s (Davis, Gallman, and Gleiter 2007). Table 9 examines what would have happened had the industry faced no such growth, or growth twice as fast. As in the petroleum exercise, it is unclear ex ante whether faster technology growth raises or lowers the welfare cost. Better technology raises the private return to harvesting, which encourages more entry and investment and worsens the externality. It also lets each ton of capacity harvest more whales, which reduces the capacity needed for a given harvest and mitigates the externality.

Laissez-faire welfare falls from \$35.46 million with zero technology growth to \$27.89 million with a growth rate of 2.7 percent. On the other hand, first-best welfare rises from \$40.02 million to \$49.62 million. As a result, the commons loss rises from 11 percent to 44 percent. With better harvesting technology, the unregulated industry builds more capacity and drives the whale stock lower. Peak vessel capacity rises from 178 to 247 thousand

TABLE 9. The cost of open access by technology growth rate

	Technology growth (per year)		
	0%	1.3% (estimated)	2.7%
<i>Panel A. Social welfare</i>			
Laissez-faire (\$M)	35.46	30.32	27.89
First best (\$M)	40.02	44.68	49.62
Difference (\$M)	4.56	14.36	21.72
Share of first best (%)	[11.4]	[32.1]	[43.8]
<i>Panel B. Laissez-faire outcomes</i>			
Lowest stock level (Million whales)	1.27	1.16	1.04
Peak capacity (Thousand tons)	178.5	224.2	246.7

Note: Each column re-solves laissez-faire and the first best with the harvesting-function trend set to the stated growth rate. In panel A, social welfare is in millions of 1880 dollars, discounted to 1815 at $\rho = 0.95$ and averaged over 300 simulations. Panel B reports the lowest stock level and the peak capacity from the laissez-faire equilibrium.

tons across the columns, and the stock's lowest level falls from 1.27 to 1.04 million whales. Technology growth is usually a source of industry profits. However, when the resource is shared, it intensifies the externality and the welfare loss grows with it. In whaling, what drove the industry's fortunes also deepened the tragedy of the commons.

6. Conclusion

This paper studies the commons problem through the lens of firm dynamics. Using firm-level panel data on the American whaling industry, I estimate a dynamic model of entry, exit, and capacity adjustment on a shared stock and use it for counterfactuals. With this framework, I evaluate the common-pool problem over the long run, where firms' entry, exit, and investment decisions shape the outcome.

Two lessons extend beyond whaling. First, valuing a shared resource requires the industry's long-run response. The shadow value of the resource depends on the capacity that will exploit it in later years. Hence, a valuation that holds the industry structure fixed misses the resource's long-run value. A policy priced on such a valuation forgoes a significant part of the attainable gain. Second, firms' responses to demand and technology determine the scale of the commons problem. Demand that lasts longer and technology that improves faster both raise the private return to exploiting the resource, a situation many modern industries face. The unregulated industry turns the higher return into entry and investment whose external cost no one pays, so the loss grows with the industry's

prospects. None of these findings is available without firm dynamics in the evaluation of the commons.

Several extensions of the current framework remain. One is to allow spatially differentiated stocks and externalities, so that the shadow value varies across grounds as well as over time. For species with distinct regional stocks, firms would choose where to hunt, which could generate a different equilibrium, though the empirical application would be more complicated. Another is to apply the framework to modern regulated industries. A harvest tax, tradable harvest rights, or restricted entry each calls for a long-run evaluation, given that most are designed within static or short-run frameworks without industry dynamics. Finally, one can apply the framework to shared-resource industries that are concentrated.³³ Dynamic imperfect competition on a shared stock has been studied in theory since Levhari and Mirman (1980) and Mason and Polasky (1997), but how market power and strategic interaction alter the tragedy of the commons over the long run is an open empirical question.

References

- Akerberg, Daniel A., Kevin Caves, and Garth Frazer. 2015. “Identification Properties of Recent Production Function Estimators.” *Econometrica* 83 (6): 2411–2451.
- Aguirregabiria, Victor, Allan Collard-Wexler, and Stephen P. Ryan. 2021. “Dynamic Games in Empirical Industrial Organization.” In *Handbook of Industrial Organization*, edited by Kate Ho, Ali Hortaçsu, and Alessandro Lizzeri, vol. 4, 225–343. Amsterdam: Elsevier.
- Aguirregabiria, Victor, and Junichi Suzuki. 2014. “Identification and Counterfactuals in Dynamic Models of Market Entry and Exit.” *Quantitative Marketing and Economics* 12: 267–304.
- Aronoff, Daniel, and Will Rafey. 2026. “Conservation Priorities and Environmental Offsets: Markets for Florida Wetlands.” *American Economic Review* 116 (5): 1723–1764.
- Aspelund, Karl M. 2025. “Redistribution in Environmental Permit Markets: Transfers and Efficiency Costs with Trade Restrictions.” Working paper.
- Association of Responsible Krill Harvesting Companies. 2026. “ARK Vision and Objectives.” <https://www.ark-krill.org/what-we-do> (accessed October 2, 2026).
- Baggio, Michele, and Metin M. Cosgel. 2024. “Racial Diversity and Team Performance: Evidence from the American Offshore Whaling Industry.” Working paper.

³³For example, about ten companies fish Antarctic krill against one catch limit, and the largest took about half of the 2025 catch (CCAMLR Secretariat 2026; Association of Responsible Krill Harvesting Companies 2026; De Augustinis 2026). Two firms draw lithium brine from the same aquifer beneath the Salar de Atacama (Flores Fernández and Alba 2023). In low-Earth orbit, where orbital space and debris form a congestible commons (Rao, Burgess, and Kaffine 2020), a single operator accounts for about two-thirds of the active satellites as of September 2026 (McDowell 2026).

- Bajari, Patrick, C. Lanier Benkard, and Jonathan Levin. 2007. "Estimating Dynamic Models of Imperfect Competition." *Econometrica* 75 (5): 1331–1370.
- Baker, C. Scott, and Phillip J. Clapham. 2004. "Modelling the Past and Future of Whales and Whaling." *Trends in Ecology & Evolution* 19 (7): 365–371.
- Benkard, C. Lanier, Przemyslaw Jeziorski, and Gabriel Weintraub. 2024. "Transitional Market Dynamics in Complex Environments." NBER Working Paper 33045.
- Bianchi, Nicola, and Michela Giorcelli. 2022. "The Dynamics and Spillovers of Management Interventions: Evidence from the Training Within Industry Program." *Journal of Political Economy* 130 (6): 1630–1675.
- Blundell, Richard, and Stephen Bond. 2000. "GMM Estimation with Persistent Panel Data: An Application to Production Functions." *Econometric Reviews* 19 (3): 321–340.
- Blundell, Richard, Stephen Bond, Michael Devereux, and Fabio Schiantarelli. 1992. "Investment and Tobin's Q: Evidence from Company Panel Data." *Journal of Econometrics* 51 (1–2): 233–257.
- Braguinsky, Serguey, Atsushi Ohyama, Tetsuji Okazaki, and Chad Syverson. 2015. "Acquisitions, Productivity, and Profitability: Evidence from the Japanese Cotton Spinning Industry." *American Economic Review* 105 (7): 2086–2119.
- Braguinsky, Serguey, Atsushi Ohyama, Tetsuji Okazaki, and Chad Syverson. 2021. "Product Innovation, Product Diversification, and Firm Growth: Evidence from Japan's Early Industrialization." *American Economic Review* 111 (12): 3795–3826.
- Brander, James A., and M. Scott Taylor. 1998. "The Simple Economics of Easter Island: A Ricardo-Malthus Model of Renewable Resource Use." *American Economic Review* 88 (1): 119–138.
- Breiwick, Jeffrey M., and Anne E. York. 2009. "Stock Assessment." In *Encyclopedia of Marine Mammals*, edited by William F. Perrin, Bernd Würsig, and J. G. M. Thewissen, 1110–1115. San Diego, CA: Academic Press.
- Butters, R. Andrew, Jackson Dorsey, and Gautam Gowrisankaran. 2025. "Soaking Up the Sun: Battery Investment, Renewable Energy, and Market Equilibrium." *Econometrica* 93 (3): 891–927.
- Caoui, El Hadi. 2023. "Estimating the Costs of Standardization: Evidence from the Movie Industry." *Review of Economic Studies* 90 (2): 597–633.
- Carter, Susan B., Scott Sigmund Gartner, Michael R. Haines, Alan L. Olmstead, Richard Sutch, Gavin Wright et al. 2006. *Historical Statistics of the United States: Millennial Edition*. vol. 3. New York: Cambridge University Press.
- CCAMLR Secretariat. 2026. "Fishery Report 2025: *Euphausia superba* in Area 48." Hobart: Commission for the Conservation of Antarctic Marine Living Resources. https://fishdocs.ccamlr.org/FishRep_48_KRI_2025.html (accessed October 2, 2026).
- Chen, Luming. 2024. "The Dynamic Efficiency of Policy Uncertainty: Evidence from the Wind Industry." Working paper.
- Clark, Colin W., Frank H. Clarke, and Gordon R. Munro. 1979. "The Optimal Exploitation of Renewable Resource Stocks: Problems of Irreversible Investment." *Econometrica* 47 (1): 25–47.

- Copeland, Brian R., and M. Scott Taylor. 2009. "Trade, Tragedy, and the Commons." *American Economic Review* 99 (3): 725–749.
- Costello, Christopher, Steven D. Gaines, and John Lynham. 2008. "Can Catch Shares Prevent Fisheries Collapse?" *Science* 321 (5896): 1678–1681.
- Costello, Christopher, Daniel Ovando, Tyler Clavelle, C. Kent Strauss, Ray Hilborn, Michael C. Melnychuk, Trevor A. Branch, Steven D. Gaines, Cody S. Szuwalski, Reniel B. Cabral et al. 2016. "Global Fishery Prospects under Contrasting Management Regimes." *Proceedings of the National Academy of Sciences* 113 (18): 5125–5129.
- Davis, Lance E., Robert E. Gallman, and Karin Gleiter. 2007. *In Pursuit of Leviathan: Technology, Institutions, Productivity, and Profits in American Whaling, 1816–1906*. Chicago: University of Chicago Press.
- Davis, Lance E., Robert E. Gallman, and Teresa D. Hutchins. 1987. "Technology, Productivity, and Profits: British-American Whaling Competition in the North Atlantic, 1816–1842." *Oxford Economic Papers* 39 (4): 738–759.
- Davis, Lance E., Robert E. Gallman, and Teresa D. Hutchins. 1988. "The Decline of US Whaling: Was the Stock of Whales Running Out?" *Business History Review* 62 (4): 569–595.
- De Augustinis, Francesco. 2026. "China and Norway Push to Increase Krill Harvests Around Antarctica." Mongabay, May 14, 2026. <https://news.mongabay.com/2026/05/china-and-norway-push-to-increase-krill-harvests-around-antarctica/> (accessed October 2, 2026).
- Elliott, Jonathan T. 2024. "Investment, Emissions, and Reliability in Electricity Markets." Working paper.
- Englander, Gabriel. 2023. "Information and Spillovers from Targeting Policy in Peru's Anchoveta Fishery." *American Economic Journal: Economic Policy* 15 (4): 390–427.
- Ericson, Richard, and Ariel Pakes. 1995. "Markov-Perfect Industry Dynamics: A Framework for Empirical Work." *Review of Economic Studies* 62 (1): 53–82.
- Flores Fernández, Cristián, and Rossella Alba. 2023. "Water or Mineral Resource? Legal Interpretations and Hydrosocial Configurations of Lithium Mining in Chile." *Frontiers in Water* 5: 1075139.
- Fowlie, Meredith, Mar Reguant, and Stephen P. Ryan. 2016. "Market-Based Emissions Regulation and Industry Dynamics." *Journal of Political Economy* 124 (1): 249–302.
- Giorcelli, Michela. 2019. "The Long-Term Effects of Management and Technology Transfers." *American Economic Review* 109 (1): 121–152.
- Gordon, H. Scott. 1954. "The Economic Theory of a Common-Property Resource: The Fishery." *Journal of Political Economy* 62 (2): 124–142.
- Gowrisankaran, Gautam, Ashley Langer, and Mar Reguant. 2026. "Energy Transitions in Regulated Markets." *American Economic Review* 116 (8): 2928–2961.
- Gowrisankaran, Gautam, Ashley Langer, and Wendan Zhang. 2025. "Policy Uncertainty in the Market for Coal Electricity: The Case of Air Toxics Standards." *Journal of Political Economy* 133

(6): 1757–1795.

- Gowrisankaran, Gautam, and Philipp Schmidt-Dengler. 2025. “A Computable Dynamic Oligopoly Model of Capacity Investment.” Working paper.
- Hanlon, W. Walker. 2015. “Necessity Is the Mother of Invention: Input Supplies and Directed Technical Change.” *Econometrica* 83 (1): 67–100.
- Hardin, Garrett. 1968. “The Tragedy of the Commons.” *Science* 162 (3859): 1243–1248.
- Hilt, Eric. 2006. “Incentives in Corporations: Evidence from the American Whaling Industry.” *Journal of Law and Economics* 49 (1): 197–227.
- Hilt, Eric. 2007. “Investment and Diversification in the American Whaling Industry.” *Journal of Economic History* 67 (2): 292–314.
- Hilt, Eric. 2008. “The Negative Trade-Off between Risk and Incentives: Evidence from the American Whaling Industry.” *Explorations in Economic History* 45 (4): 424–444.
- Ho, Phuong. 2023. “Why Is Trading So Important in Cap and Trade? The Role of Economies of Scale and Productivity.” Working paper.
- Homer, Sidney, and Richard Eugene Sylla. 1996. *A History of Interest Rates*. New Brunswick, NJ: Rutgers University Press.
- Hopenhayn, Hugo A. 1992. “Entry, Exit, and Firm Dynamics in Long Run Equilibrium.” *Econometrica* 60 (5): 1127–1150.
- Hornbeck, Richard, Shanon Hsuan-Ming Hsu, Anders Humlum, and Martin Rotemberg. 2025. “Gaining Steam: Technology Diffusion with Recurring Lock-In.” Working paper.
- Huang, Ling, and Martin D. Smith. 2014. “The Dynamic Efficiency Costs of Common-Pool Resource Exploitation.” *American Economic Review* 104 (12): 4071–4103.
- Juhász, Réka. 2018. “Temporary Protection and Technology Adoption: Evidence from the Napoleonic Blockade.” *American Economic Review* 108 (11): 3339–3376.
- Juhász, Réka, Mara P. Squicciarini, and Nico Voigtländer. 2024. “Technology Adoption and Productivity Growth: Evidence from Industrialization in France.” *Journal of Political Economy* 132 (10): 3215–3259.
- Kaiser, Brooks A. 2013. “Bioeconomic Factors of Natural Resource Transitions: The US Sperm Whale Fishery of the 19th Century.” Working paper.
- Kalouptzidi, Myrto. 2018. “Detection and Impact of Industrial Subsidies: The Case of Chinese Shipbuilding.” *Review of Economic Studies* 85 (2): 1111–1158.
- Lamoreaux, Naomi R. 1994. *Insider Lending: Banks, Personal Connections, and Economic Development in Industrial New England*. Cambridge: Cambridge University Press.
- Lane, Nathan. 2025. “Manufacturing Revolutions: Industrial Policy and Industrialization in South Korea.” *Quarterly Journal of Economics* 140 (3): 1683–1741.
- Levhari, David, and Leonard J. Mirman. 1980. “The Great Fish War: An Example Using a Dynamic Cournot-Nash Solution.” *Bell Journal of Economics* 11 (1): 322–334.
- Levinsohn, James, and Amil Petrin. 2003. “Estimating Production Functions Using Inputs to Control

- for Unobservables.” *Review of Economic Studies* 70 (2): 317–341.
- Libecap, Gary D. 2024. “Variation in State Response to Open Access: Constituent Characteristics.” NBER Working Paper 32021.
- Mason, Charles F., and Stephen Polasky. 1997. “The Optimal Number of Firms in the Commons: A Dynamic Approach.” *Canadian Journal of Economics* 30 (4): 1143–1160.
- McDowell, Jonathan C. 2026. “Active Satellites in Orbit.” Jonathan’s Space Report. <https://planet4589.org/space/stats/active.html> (accessed October 2, 2026).
- Moment, David. 1957. “The Business of Whaling in America in the 1850’s.” *Business History Review* 31 (3): 261–291.
- Mystic Seaport Museum, New Bedford Whaling Museum, and Nantucket Historical Association. 2024a. “American Offshore Whaling Crew Lists.” Mystic, CT: Mystic Seaport Museum. <https://whalinghistory.org/av/crew/about/> (accessed September 23, 2025).
- Mystic Seaport Museum, New Bedford Whaling Museum, and Nantucket Historical Association. 2024b. “American Offshore Whaling Voyages: A Database.” Mystic, CT: Mystic Seaport Museum. <https://whalinghistory.org/av/> (accessed September 23, 2025).
- Nicholas, Tom. 2019. *VC: An American History*. Cambridge, MA: Harvard University Press.
- Noack, Frederik, and Christopher Costello. 2024. “Credit Markets, Property Rights, and the Commons.” *Journal of Political Economy* 132 (7): 2396–2450.
- Ostrom, Elinor. 1990. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge: Cambridge University Press.
- Rao, Akhil, Matthew G. Burgess, and Daniel Kaffine. 2020. “Orbital-Use Fees Could More Than Quadruple the Value of the Space Industry.” *Proceedings of the National Academy of Sciences* 117 (23): 12756–12762.
- Rubens, Michael. 2023. “Management, Productivity, and Technology Choices: Evidence from US Mining Schools.” *RAND Journal of Economics* 54 (1): 165–186.
- Rust, John. 1987. “Optimal Replacement of GMC Bus Engines: An Empirical Model of Harold Zurcher.” *Econometrica* 55 (5): 999–1033.
- Ryan, Stephen P. 2012. “The Costs of Environmental Regulation in a Concentrated Industry.” *Econometrica* 80 (3): 1019–1061.
- Segerson, Kathleen, and Dale Squires. 1993. “Capacity Utilization under Regulatory Constraints.” *Review of Economics and Statistics* 75 (1): 76–85.
- Smith, Vernon L. 1969. “On Models of Commercial Fishing.” *Journal of Political Economy* 77 (2): 181–198.
- Stavins, Robert N. 2011. “The Problem of the Commons: Still Unsettled after 100 Years.” *American Economic Review* 101 (1): 81–108.
- Taylor, M. Scott. 2011. “Buffalo Hunt: International Trade and the Virtual Extinction of the North American Bison.” *American Economic Review* 101 (7): 3162–3195.
- Tillman, Michael F., and Gregory P. Donovan. 1983. *Historical Whaling Records: Including the Proceed-*

- ings of the International Workshop on Historical Whaling Records, Sharon, Massachusetts, September 12–16, 1977. Reports of the International Whaling Commission, Special Issue 5. Cambridge, UK: International Whaling Commission.*
- Tower, Walter Sheldon. 1907. *A History of the American Whale Fishery*. Philadelphia: University of Pennsylvania.
- United Nations. 2014. *System of Environmental-Economic Accounting 2012: Central Framework*. New York: United Nations.
- Weintraub, Gabriel Y., C. Lanier Benkard, and Benjamin Van Roy. 2008. “Markov Perfect Industry Dynamics with Many Firms.” *Econometrica* 76 (6): 1375–1411.
- Weintraub, Gabriel Y., C. Lanier Benkard, and Benjamin Van Roy. 2010. “Computational Methods for Oblivious Equilibrium.” *Operations Research* 58 (4, pt. 2): 1247–1265.
- Whitehead, Hal. 2002. “Estimates of the Current Global Population Size and Historical Trajectory for Sperm Whales.” *Marine Ecology Progress Series* 242: 295–304.